

Silent order corruption in per-sample evaluation artefacts: control degeneration, diagnostic blind spots, and a provenance contract

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ABSTRACT

Per-sample error tables, difficulty analyses, sample-level ablations and per-instance routing all join two artefacts written by different parts of a pipeline, and they join them positionally. In a widely used forecasting library one default—validation shuffles, test does not—turns a per-window artefact taken from the validation split into a permutation of the intended vector, and the fault survives the usual guards: aggregate metrics and model selection are exactly invariant, and the vectors are equally long. Our central result is structural. Under uniform shuffling the permuted table is *equal in distribution* to the table produced by the canonical shuffled-feature negative control for every order-symmetric statistic, so a null validated by that control is as consistent with a broken join as with a true null; when the label is derived from several independently written vectors, the control instead preserves the label law and no longer bounds the damage. Two post-hoc diagnostics recover part of the discarded order—an autocorrelation order test with exact randomisation moments, and a cross-arm agreement test—but on 356 single-GPU runs we map blind spots rather than only power: the first flags 90.63% of arms in the real defect yet at most 1.67% of injected block and partial permutations short of full randomisation, and the second is defeated when two arms share a loader order, as 20 of 24 real defective groups do. The primary defence is therefore write-side: a provenance contract that binds each array to an ordered sample-identifier digest and fails closed stops 20 of 25 injected identity faults, and we enumerate the 5 it misses. Downstream the fault erases a real association while the headline metric stays bit-identical, and the one-line repair moves reported test error by up to 4.316%. Contract, diagnostics and all artefacts are released.

1. Introduction

Empirical machine learning has moved past the aggregate number. A modern paper does not only report mean squared error on a test split; it reports which samples are hard and why [1], fits a model of per-sample difficulty, routes each sample to one of several arms, correlates per-sample loss with per-sample covariates, and ablates features against a per-sample outcome. Every one of those analyses consumes a *per-sample artefact*: a vector with one entry per evaluation sample, written by the evaluation loop, and later joined against a second table with one row per evaluation sample, written by a different piece of code—often a different process, a different day, a different author.

The join is almost always *positional*. Entry i of the error vector is assumed to describe row i of the feature table. Positional joins are the default in every array-oriented stack [2, 3]: a shape check is one line and a key-based join needs a key that nobody wrote down. This paper is about what happens when the assumption is wrong—when the two artefacts were written in different orders—and about why that particular fault is unusually hard to find.

The carrier we use is concrete and widespread. In the Time-Series-Library family of long-horizon forecasting codebases [4, 5], which is the de facto reference implementation for a large fraction of the recent literature [6, 7, 8, 9, 10], the data provider constructs the validation loader with shuffling enabled and the test loader with shuffling disabled. That asymmetry is harmless for its intended purpose: the validation split is consumed only through a mean. It stops being harmless the moment anyone stores a *per-window* quantity

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from the validation split—which is exactly what per-window difficulty analysis, window-level gating and per-sample ablation require. The stored vector is then in loader order, the feature table is in window order, and the positional join silently pairs window i 's features with a uniformly random other window's error.

What makes this fault worth a paper is not that a permutation destroys information—that is obvious—but the structure of its invisibility. We prove three things about it. First, the fault is invariant to every aggregate the pipeline actually checks: the validation mean, the early-stopping trajectory, the model-selection decision and the headline table are bit-identical, because a mean is a function of a multiset (Proposition 2). Second, the length check that array code performs instead of a key check is blind by construction whenever the last partial batch is kept, which is the standard setting for evaluation loaders (Proposition 3). Third—and this is our main conceptual contribution—the corrupted joined table is *equal in distribution* to the table produced by the negative control that the community uses to validate exactly this kind of analysis, namely shuffling the feature rows (Proposition 5). A shuffled-feature control is meant to show that a reported association is not an artefact of the pipeline. Under the defect it shows nothing at all: the “real” arm and the control arm are the same experiment, so the control cannot fail. Any statistic that reads the joined table as an exchangeable set of pairs—Spearman or Pearson correlation, decile contrasts, R^2 of a fitted difficulty model, permutation feature importance, per-sample gating accuracy—inherits the degeneracy.

The consequence is asymmetric and, we argue, under-appreciated. A broken join cannot manufacture a positive finding: it attenuates every association towards the null (Proposition 4). It can, and in our experiments does, manufacture a *negative* finding, and negative findings about per-sample structure are published: “window difficulty is not predictable”, “per-sample routing does not help”, “these features carry no signal”. Such a claim, accompanied by a shuffled-feature control that behaves “as expected”, is indistinguishable from a silent order bug. We wrote this paper because we hit the fault ourselves while auditing per-window plug-in selection [11], where the conclusion under study was precisely of that shape.

The situation is not hopeless, but the recoverable margin is narrower than the fault's simplicity suggests. The aggregate discards the information needed to detect the fault; the artefact itself retains some of it. Two structures survive in a per-sample error vector and are weakened by a permutation. (i) In sliding-window forecasting, consecutive windows overlap in $L + H - 1$ of their $L + H$ time steps, so consecutive per-window errors are strongly dependent; the intact vector therefore has a serial correlation near one, and a uniformly permuted vector has none. We turn this into the *autocorrelation order test* (AOT), whose randomisation distribution has exact first and second moments [12], hence an $O(n)$ p -value with no resampling. (ii) When several arms are evaluated on the same samples, their per-sample errors agree strongly because sample difficulty dominates arm identity; if a defective pipeline draws an *independent* permutation per arm, the agreement collapses. That is the *cross-arm agreement test* (CAT), which needs no serial structure and therefore applies to independent and identically distributed data, where AOT has no signal. Neither test needs the model, the data or a rerun—they read the published vectors, so a third party can audit an artefact they merely downloaded.

Our empirical contribution about these two diagnostics is mostly negative, and we report it that way because it is what the data support. Both tests have sharp operating boundaries, and in this study both boundaries turned out to be inside the space of faults that a real pipeline produces. AOT keys on the loss of serial dependence, so it flags a near-uniform permutation—90.63% of arms on our real defective artefacts—but it does not flag a corruption that preserves long contiguous runs: over injected block permutations at four block lengths its flag rate is 0%, and for partial permutations of 5% and 10% of positions it is 0%. CAT keys on the collapse of cross-arm agreement, which presumes that each arm was permuted independently. We audited that presumption against our own archive instead of assuming it, and it does not hold: of 24 real defective four-arm groups, only 2 have pairwise distinct loader orders, while 20 share at least one order between two arms and 2 share one order across all four. Under exactly that sharing structure CAT certifies 22 of 24 genuinely corrupted groups as clean. A controlled sweep confirms the mechanism: with independently drawn permutations CAT flags 24 of 24 groups, and with even a single shared pair among four arms it flags none. The honest reading is that AOT is a targeted diagnostic for near-random order corruption and CAT requires independently produced orders; neither is a general integrity check.

That is why the primary defence has to be a write-side one, and why we spend more of this paper on it than the original framing would suggest. A sidecar that merely names the order an array is in is not enough—an equally long array from a different sample set, an array rotated by one, or a stale sidecar all

survive such a check. We therefore specify a contract that binds each array to a digest of its *ordered* sample identifiers, a digest of its own values, and the identity of the sample set it came from, and that fails closed when any of those disagree. We implement it, back-fill it over 356 archived cells, and test it adversarially: it stops 20 of 25 injected faults, including rotation, arbitrary permutation, stale sidecars and equally long arrays from a different horizon. It does not stop 5, and we enumerate them, because a contract whose limits are unstated invites exactly the misplaced confidence this paper is about.

Finally, we quantify a practical trap in the repair. Removing the shuffle from a validation loader changes how many pseudo-random numbers the run consumes and therefore the whole subsequent training trajectory. Across 96 matched pairs that differ only in this flag, reported test error moves by a median of 0.019% and a maximum of 4.316%, and the selected epoch changes in a substantial fraction of pairs. A maintainer who fixes the flag and compares against previously published numbers is therefore comparing two different experiments, and must re-run rather than patch.

Contributions

1. **A named fault class with an exact invisibility argument.** We formalise *silent order corruption*: a positional join between two per-sample artefacts written in different iteration orders. We prove that it is invariant to all order-symmetric aggregates (Proposition 2), that the standard length check cannot see it (Proposition 3), and that it attenuates associations monotonically towards the null (Proposition 4).
2. **Control degeneration.** We prove that the corrupted table and the canonical shuffled-feature negative control are equal in distribution (Proposition 5), so the control that is supposed to protect a per-sample claim is, for this fault, uninformative by construction: no statistic of the joined table can separate the two arms. To our knowledge this failure mode of negative controls has not been stated. We verify the equality empirically with equivalence tests rather than with a failure to reject (Section 7.5). We also identify the one case where the two differ—analyses whose label is derived from several independently written vectors, such as per-sample arm selection—and show that the difference runs the wrong way: the control preserves the label law and therefore reports a *milder* failure than the defect causes (Proposition 7).
3. **Two post-hoc order diagnostics, and a measurement of where they stop working.** AOT, with exact randomisation moments and a closed-form power expression, and CAT, for data without serial structure. We calibrate both against Monte-Carlo nulls, and we then spend most of the evaluation on their boundaries rather than on their power: the corruption severities AOT does not flag, the cross-arm order-sharing patterns that defeat CAT, and—using the observed permutations in our own archive rather than simulated ones—how often those adverse conditions actually occur (Sections 4, 7.2). We also correct the analytic reference distribution for CAT, whose earlier form omitted the $\binom{K}{2}$ pairs being averaged and was consequently mis-scaled by 0.4098 (Section 4.2).
4. **A provenance contract that verifies sample identity, not just an order label,** with a reference implementation, a write-side and read-side API, a back-fill over the archived artefacts, and an adversarial test suite. We report a machine-generated coverage table listing both the 20 faults the contract stops and the 5 it does not (Section 5).
5. **An empirical study** of 356 single-GPU runs over 7 datasets, 3 backbones, 4 arms and four horizons (21.3 GPU hours), which (a) confirms on real artefacts that the permutation is exactly a permutation and the aggregate is unchanged to machine precision, (b) measures detector operating characteristics and blind spots on 712 real sequences covering 4 265 096 window-level evaluations, (c) shows a real per-window association collapsing from Spearman 0.2192 to -0.0004 under the defect, using dependence-aware and cluster-aware inference rather than an independent-pairs approximation, and (d) quantifies the cost of the repair.
6. **Actionable guidance** for authors, reviewers and artefact evaluators, in the form of a checklist tied to the contract and, where a diagnostic is applicable, to the tests (Section 8).

What we do not claim We do not claim that any published result is wrong. We did not audit other people’s artefacts; we characterise a fault we found in our own pipeline, built on a public library whose defaults make the fault easy to reach, and we quantify what it does. We do not claim that the two diagnostics detect order corruption in general. They were not observed to detect block permutations or mild partial permutations

under the protocol we evaluated, and CAT was not observed to detect any corruption once two arms shared a loader order; what they certify is the absence of a specific, near-uniform scrambling, on artefacts of the lengths we studied, and Section 9 lists what they cannot see. We do not claim that the contract certifies correctness: it certifies which sample each number belongs to, never that the number is the right statistic. We do not claim that the fault is confined to time series; the formal statements only require a per-sample artefact and a positional join, and sliding windows merely make one of the two diagnostics unusually powerful. All statistical statements below are calibrated at the sample sizes we evaluated; where a claim rests on the 7 datasets in the grid rather than on the 356 runs, we report it with the datasets as the resampling unit and accept the resulting wide intervals.

2. The fault

2.1. Setting and notation

Fix an evaluation split with n samples. In the sliding-window forecasting setting that we use throughout, sample j is the window that starts at time j : it consists of a look-back of L steps and a target of H steps, and $j \in \{0, \dots, n-1\}$ enumerates the windows in calendar order. Write $\ell = (\ell_0, \dots, \ell_{n-1})$ for the per-sample loss, ℓ_j being the mean squared error of the trained model on window j , and $X \in \mathbb{R}^{n \times d}$ for a table of per-sample covariates, row j describing window j . In our experiments X holds $d = 10$ cheap descriptors of the look-back window only (volatility, range, mean absolute increment, trend slope, lag-1 autocorrelation, level drift, spectral entropy, cross-channel correlation, deviation of the last point from the window mean, kurtosis); nothing in the analysis depends on the particular choice.

Definition 1 (Index order and loader order). *An array of length n is in index order if its i -th entry describes sample i . It is in loader order if its i -th entry describes the sample visited i -th by the data loader that produced it. The two coincide if and only if the loader is sequential.*

A data loader that shuffles draws a permutation $\sigma \in S_n$ and visits samples $\sigma(0), \sigma(1), \dots$. An evaluation loop that appends one number per batch element therefore writes the vector

$$a_i = \ell_{\sigma(i)}, \quad i = 0, \dots, n-1, \quad (1)$$

which is in loader order. A downstream consumer that joins positionally against X believes $a_i = \ell_i$. We call the resulting analysis *order corrupted*, and we call the fault *silent* because, as Section 2.3 shows, nothing the pipeline checks changes.

2.2. How the fault is reached in practice

The fault needs two ingredients that live far apart in the source: a shuffling evaluation loader, and a consumer that joins positionally. Listing 1 shows the first as it appears in the Time-Series-Library data provider [4], quoted verbatim from the copy vendored in our replication package (SHA-256 8034b4a9...ad9a).

Listing 1: The provider that makes the fault reachable, quoted verbatim. Only `test` is protected, and the last partial batch is kept for every split. The flag is correct for its intended purpose—the validation split is consumed through a mean—and becomes a hazard the moment a per-sample artefact is stored from the validation split.

```

1 def data_provider(args, flag):
2     ...
3     shuffle_flag = False if (flag == 'test' or flag == 'TEST') else True
4     drop_last = False      # kept for every split, including 'val'
5     ...
6     data_loader = DataLoader(data_set, batch_size=batch_size,
7                             shuffle=shuffle_flag,      # 'val' -> True
8                             num_workers=args.num_workers,
9                             drop_last=drop_last)

```

Two properties of this snippet matter. First, the asymmetry is invisible at the call site: the caller asks for `data_provider(args, 'val')` and receives a loader whose order is a global, undocumented policy decision expressed as a one-line ternary. Second, the last partial batch is kept for every split (`drop_last=False`), so an evaluation loop adapted to save per-sample quantities writes exactly one entry per validation window, whatever the order. Nothing has to be modified for the last accidental guard to be absent: the length of the saved vector already matches the feature table under both orders, as Proposition 3 makes precise.

Listing 2 shows the consumer. It is four lines, it contains a shape assertion, and it is wrong.

Listing 2: The consumer. The assertion passes, the join is wrong, and no exception is ever raised.

```

1 err = np.load(f"{run}/val_window_mse.npy") # loader order (unknown to the caller)
2 feat = build_window_features(dataset, split="val") # index order
3 assert len(err) == len(feat) # passes: both equal n
4 rho = spearmanr(feat["volatility"], err).statistic # meaningless
    
```

We stress that neither listing is unusual, and that the fault is not a bug in the library: the library never promised an order, and never stored a per-sample artefact from the validation split. The fault is created by the composition, which is precisely why it survives code review of either half in isolation. This is the shape of fault that the software-engineering literature calls *silent*: it produces no crash, no warning and no visibly wrong output [13, 14, 15].

2.3. Why the pipeline’s own guards cannot see it

Proposition 2 (Aggregate invariance). *Let T be any functional of a vector that depends only on its multiset of entries, and let a be as in (1). Then $T(a) = T(\ell)$ for every $\sigma \in S_n$. In particular the validation mean, every quantile, the histogram, and hence the early-stopping trajectory, the selected checkpoint and every order-symmetric aggregate computed from the validation split are unchanged. Aggregates that are not order-symmetric—a per-window plot, a rolling statistic, or any join keyed on position—are of course affected; the point of the proposition is that these are not the quantities a training pipeline checks.*

PROOF. a is a rearrangement of ℓ , so the two share a multiset of entries, and T depends on nothing else. Early stopping compares scalar aggregates across epochs; if each epoch’s aggregate is unchanged, the comparison is unchanged, and so is the resulting checkpoint. $\square$

Proposition 2 is the reason the fault is not caught by the experiment itself. Every number that appears in a conventional results table is computed from a multiset. The fault lives entirely in the *index* of the artefact, and conventional reporting never reads the index.

Proposition 3 (Blindness of the length check). *Let the loader use batch size b and let $n = qb + r$ with $0 \leq r < b$. If the loader keeps the last partial batch, the saved vector has length n for every σ , so a length or shape assertion against X passes identically under the defect and under the intended behaviour. If instead the loader drops the last batch, the saved length is qb under both orders; the assertion then fails—but it fails for a reason unrelated to order, and the standard repair is to keep the last batch, which restores blindness.*

PROOF. Immediate: the number of visited samples does not depend on the visiting order. $\square$

The practical reading of Proposition 3 is uncomfortable. The one guard that array-oriented code reliably writes—a shape assertion—is exactly the guard that this fault is immune to, and the immunity is strongest in the configuration that a careful author will choose.

2.4. What the fault does to inference

Proposition 4 (Attenuation to the null). *Let σ be drawn uniformly from S_n independently of (X, ℓ) , and let $\hat{\rho}$ be the sample Spearman correlation between a column of X and the joined vector a . Then, conditionally on (X, ℓ) ,*

$$\mathbb{E}[\hat{\rho}] = 0, \quad \text{Var}[\hat{\rho}] = \frac{1}{n-1}, \quad (2)$$

so $\hat{\rho} = O_P(n^{-1/2})$ regardless of the true association, and any test of $\hat{\rho} = 0$ has size equal to its nominal level. The same holds for any rank-based or moment-based association statistic that is a function of the joined pairs and has population value zero under independence.

PROOF. Under a uniform σ , the ranks of a form a uniformly random permutation independent of the ranks of the covariate column; (2) is the classical randomisation mean and variance of Spearman's coefficient under independence. The statement for other statistics follows because the joined pairs are exchangeable with independent margins. $\square$

Proposition 4 says the fault is *conservative for positive claims*: it cannot fabricate an association, because it destroys the pairing that any association would have to be carried by. It is exactly the opposite of a leakage fault, which inflates results [16]. That asymmetry has been read as reassurance in discussions of pipeline hygiene. It is not reassurance. It means the fault specialises in producing the one kind of finding that no positive-result check will catch: a null.

3. Control degeneration: why the standard negative control cannot fail

When a paper reports a per-sample analysis, the accepted way to show that the analysis is not an artefact of the pipeline is a negative control [17]: re-run it with the association deliberately destroyed and check that the reported effect disappears. In per-sample machine learning the canonical instance is the *shuffled-feature control*—permute the rows of X , keep ℓ , refit, and verify that the signal vanishes; it is the sample-level analogue of the random-label control used to probe memorisation. It is cheap, it is standard, and against this fault it is useless.

Proposition 5 (Distributional equality of the defect and the control). *Let σ, π be independent and uniform on S_n and independent of (X, ℓ) . Define the defect table $D = \{(X_i, \ell_{\sigma(i)})\}_{i=0}^{n-1}$ and the shuffled-feature control table $C = \{(X_{\pi(i)}, \ell_i)\}_{i=0}^{n-1}$. Then D and C have the same distribution as multisets of pairs. Consequently, for every statistic S that is a function of the multiset of joined pairs,*

$$S(D) \stackrel{d}{=} S(C). \quad (3)$$

PROOF. Re-index C by $j = \pi(i)$: as a multiset, $C = \{(X_{\pi(i)}, \ell_i)\}_i = \{(X_j, \ell_{\pi^{-1}(j)})\}_j$. Since π is uniform on S_n , so is π^{-1} , and π^{-1} is independent of (X, ℓ) . Hence C has the law of $\{(X_j, \ell_{\tau(j)})\}_j$ with τ uniform, which is the law of D . A statistic that reads only the multiset of pairs is a measurable function of it, so its law is determined. $\square$

Corollary 6 (The control cannot discriminate). *Let S be a statistic that depends on the joined table only through its multiset of pairs. Then no test that compares S computed on the joined table against S computed on the shuffled-feature control has power against silent order corruption; its power equals its size for every n and every (X, ℓ) .*

The invariance requirement in Corollary 6 is a real restriction, and stating it precisely is what makes the result usable. A statistic is covered if relabelling the rows of the joined table leaves it unchanged. That covers most of what per-sample sections actually report: Pearson and Spearman correlations, mutual information estimates, decile or quintile contrasts of the loss across a covariate, permutation feature importance, per-sample calibration curves, per-window classification of the best arm, and the test-split performance of a model fitted on *all* rows of the joined table. Each of these treats the joined table as an exchangeable collection of pairs—the reason they are legitimate in the first place—and that is exactly what makes them blind here.

The requirement genuinely excludes one family, and we flag it because it is easy to assume the result is more general than it is. Any statistic that reads a row's *position* is not a multiset function, and for such a statistic the defect and the control need not agree. The practical instance is a split defined by position: a chronological or purged train/holdout split of the joined table [18], blocked cross-validation, or any “first half versus second half” contrast. Under the defect the loss values are permuted globally, so both halves receive a

random sample of losses; under the shuffled-feature control the losses stay in place, so the second half keeps the losses that genuinely belong to the later period, and the two arms can differ systematically whenever the loss level drifts across the split. Section 7.5 measures this on our grid: the purged-holdout statistic is the one statistic in the study where defect and control separate detectably, in the direction the argument predicts. This is not a weakness of the theory but a boundary of it—and a narrow one, because an author who chose a position-based holdout to validate a per-sample claim has an accidental partial guard, not a designed one: it responds to drift in the loss level, not to the corruption itself, so it fires on some pipelines and stays silent on others.

Three points deserve emphasis.

The degeneracy is not asymptotic. The two laws are equal for every n , so no amount of data helps, and a larger evaluation split makes the null look *cleaner*, because both arms converge to the same null value at rate $n^{-1/2}$.

It survives multiplicity correction. Reporting a family of features with a false-discovery-rate correction [19] controls the error rate of the family under the null, which is precisely the situation the defect creates. The correction behaves impeccably and confirms the wrong conclusion.

It survives equivalence testing. If an author is careful enough to substantiate a null with a two one-sided test procedure [20, 21] rather than with a failure to reject, the conclusion strengthens: they will correctly certify that the observed effect lies inside their equivalence margin, because it genuinely does—for the corrupted table. We use this in reverse in Section 7.5: we certify the equality predicted by Proposition 5 instead of merely failing to reject it.

3.1. One exception, and it makes the control worse

Proposition 5 assumes the analysis consumes a single loss vector. Some per-sample analyses instead derive their target from several vectors that are produced by *separate* runs—per-sample arm selection is the common case, where the label of window i is the index of the arm with the smallest loss on that window. A defective pipeline permutes each of those vectors independently, because each arm is a separate process with its own loader, and that is a different object from a single permutation.

Proposition 7 (Derived labels: the control keeps the label law, the defect does not). *Fix K arms evaluated on the same n windows, let $\ell^{(k)}$ denote the index-order loss vector of arm k , and let $A_i = \arg \min_k \ell_i^{(k)}$ be the derived label. Let the permutations $\sigma_1, \dots, \sigma_K$ and π be independent and uniform on S_n .*

1. *Under the shuffled-feature control the label vector is $(A_i)_i$ itself, joined to the π -permuted feature rows; its law, and in particular the marginal $\mathbb{P}(A_i = k)$, is unchanged.*
2. *Under the defect the label is $\tilde{A}_i = \arg \min_k \ell_{\sigma_k(i)}^{(k)}$. For each fixed i the K entries $\ell_{\sigma_k(i)}^{(k)}$ are independent draws from the K empirical marginals, so $\tilde{A}_i$ is distributed as the arg min under the product of those marginals, not under their joint law.*

Consequently D and C are in general not equal in distribution for derived-label analyses: the control preserves the label law exactly, the defect replaces it by the arg min law of independent coordinates.

PROOF. Part 1 is immediate: the control does not touch ℓ . For part 2, fix i ; each $\sigma_k(i)$ is uniform on $\{0, \dots, n-1\}$ and the σ_k are independent, so the vector $(\ell_{\sigma_k(i)}^{(k)})_k$ has the product law of the K empirical distributions, whereas $(\ell_i^{(k)})_k$ is a draw from their joint law. Window difficulty is shared across arms, so the joint law is far from the product law, and the arg min functional distinguishes them. $\square$

Two remarks on what Proposition 7 does and does not say. It does not rescue the control: the control still has no power against the defect, because a null remains a null. What it removes is the consolation that the control is at least a *conservative* stand-in. Since the control preserves the label marginal, it also preserves the fallback of predicting the globally best arm, whereas the defect dilutes that marginal towards the product-law arg min; an author reading the control therefore sees a milder failure than the defect actually causes. Which of the two label laws is flatter is not determined by the proposition alone—that depends on the joint law of the arms—so we treat the *direction* as an empirical question and measure it in Section 7.5,

where the modal-arm mass falls from 46.1% under the correct join to 33.3% under the defect while the control leaves it untouched by construction.

The escape route is therefore not a better control on the joined table. It is to test the artefact's *order* directly, before any join, using structure that the aggregate discards. That is the subject of the next section.

4. Two post-hoc order diagnostics

We now build diagnostics that read a stored per-sample vector and ask whether it retains the structure an index-ordered vector should have. The design constraints are practical: no access to the model, the data or a rerun; a cost of at most $O(n)$ or a few seconds of resampling; and known error rates, because a diagnostic without a calibrated false-positive rate will be ignored. Because the direction of the test is a recurring source of confusion—including in an earlier version of this manuscript—we fix terminology before stating anything else.

Terminology Each diagnostic tests a null hypothesis of *complete randomisation*. We use three terms and never interchange them.

- *Certified structured order*: the null of complete randomisation was rejected ($p < \alpha$). The vector retains the dependence structure that an intact vector has. This is evidence *against* uniform scrambling; it is not proof of alignment, because many misalignments preserve structure.
- *Not certified*: the null was not rejected. There is insufficient evidence of structured order. This is not proof that the vector was scrambled: a short vector, a weakly dependent loss, or a genuinely aligned but low-autocorrelation artefact all fail to certify.
- *Corruption detected*: the diagnostic failed to certify a vector that, in a controlled experiment, we know to be corrupted. This is an evaluation term. It is only available when the fault label is known, and it must never be used to describe a verdict on an artefact of unknown provenance.

The practical consequence is asymmetric and worth stating plainly. A certified vector may still be misaligned. An uncertified vector may still be correct. What the diagnostics buy is a cheap, high-signal alarm on one specific failure mode, not an integrity guarantee.

4.1. AOT: the autocorrelation order test

Why there is signal at all Consecutive sliding windows share $L + H - 1$ of their $L + H$ time steps. Their targets therefore overlap almost completely, and so do the local regimes that determine how hard they are to predict. The per-sample loss of an intact vector is consequently a strongly, positively dependent sequence—empirically, a lag-1 correlation of at least 0.85 in every one of the 712 sequences we measured (Section 7.2). A permutation destroys that dependence completely. Order is not recoverable from the multiset, but the *presence of serial dependence* is a one-bit summary of order, and one bit is all we need.

Statistic Let $x \in \mathbb{R}^n$ be the stored vector, $c = x - \bar{x}$, and let indices be read modulo n . Define the circular serial correlation

$$r(x) = \frac{\sum_{i=0}^{n-1} c_i c_{i+1}}{\sum_{i=0}^{n-1} c_i^2}. \quad (4)$$

We use the circular rather than the linear form for one reason: under the randomisation null the circular statistic has *exact* first and second moments, which removes the need to resample [12].

Null and p-value The null hypothesis is H_0 : “the stored vector is a uniformly random rearrangement of its own values”, which is the law of the artefact under the defect (1). Note that this null is nonparametric and conditions on the observed multiset, so it makes no distributional assumption about the loss. With power sums $S_k = \sum_i c_i^k$ (so $S_1 = 0$),

$$\mathbb{E}_{H_0}[r] = -\frac{1}{n-1}, \quad (5)$$

$$\text{Var}_{H_0}[r] = \frac{1}{S_2^2} \left[\frac{S_2^2 - S_4}{n-1} + \frac{S_2^2 - 2S_4}{(n-1)(n-2)} - \frac{S_2^2}{(n-1)^2} \right], \quad (6)$$

and r is asymptotically normal under H_0 [12]. Writing $z = (r - \mathbb{E}_{H_0}[r])/\sqrt{\text{Var}_{H_0}[r]}$, we report the one-sided p -value $p = 1 - \Phi(z)$, one-sided because the alternative created by intact overlapping windows is positive dependence. Evaluating (4)–(6) costs one pass over the vector; on the longest sequences in our study ($n = 11\,425$) the test runs in microseconds. For calibration we also compute a Monte-Carlo p -value by resampling permutations, and Section 7.2 shows the two agree to 0.0262 in the worst case over 120 sequences.

Certification rule and power Fix a level α . We say a vector is *certified in structured order* at level α if $p < \alpha$, i.e. if the complete-randomisation null is rejected. On a vector known to be corrupted, $p < \alpha$ therefore means the diagnostic *failed* to flag the corruption. Because $\text{Var}_{H_0}[r] \approx n^{-1}$ up to a kurtosis correction, the serial correlation that the test detects with probability $1 - \beta$ is

$$\rho_{1-\beta}(n, \alpha) \approx \frac{z_\alpha + z_\beta}{\sqrt{n}}, \quad \text{power}(\rho) \approx \Phi(\rho\sqrt{n} - z_\alpha), \quad (7)$$

with z_α the upper- α normal quantile. Two consequences are worth stating because they bound the applicability of the test. Sequences as short as a few hundred samples suffice when the dependence is strong: at $\alpha = 0.05$ and 90% power the detectable ρ is 0.2926 at $n = 100$ and 0.0925 at $n = 1000$, both far below what window overlap produces. And the test degrades gracefully rather than abruptly: as overlap decreases—longer stride, subsampled windows, independent samples— ρ falls and the test loses power smoothly, which is why we pair it with a second test that does not depend on serial structure at all.

Equation (7) is a power expression against the alternative “exchangeable values”, and that is the only alternative it covers. A corruption that leaves long contiguous runs intact—a block permutation, a partial permutation of a few per cent of positions, an off-by-one shift—does not move ρ appreciably, so (7) says nothing about it and the test has essentially no power against it. Section 7.2 measures this directly rather than leaving it as a caveat, and the measurement is unfavourable.

How deep the normal tail may be read The moments (5)–(6) are exact, but the normal reference is not, so a p -value quoted at 10^{-6} is an extrapolation of a two-moment approximation and has to be checked rather than trusted. Per-window squared errors are heavy tailed—sample kurtosis reaches 14.1 in our bank—so the randomisation law of r need not have Gaussian tails. We measured the discrepancy instead of assuming it away. On 5 sequences stratified by length we drew 500 000 independent uniform permutations of each, which is exactly the null of the test and exactly the law of the artefact under the defect, and counted how often the normal rule rejects (Table 1, 2 500 000 null draws in total). The rule is close to nominal at $\alpha = 10^{-2}$ (worst ratio $1.26\times$) and progressively *anti-conservative* deeper in the tail: up to $1.77\times$ nominal at 10^{-3} , $2.3\times$ at 10^{-4} and $2.8\times$ at 10^{-5} ; at 10^{-6} the largest observed ratio is $10\times$ nominal, on exceedance counts small enough that the ratio itself is poorly resolved at this B . Across the 20 sequence-by-level cells we measured, 80% are anti-conservative. Levels below 10^{-3} must therefore be read as thresholds on z , not as calibrated error probabilities, and we do not attach an exact guarantee to any of them: what we report is an empirical calibration at the sample sizes and permutation budget evaluated here.

Two facts keep this from undermining the certification rule, and both are measured rather than assumed. First, the approximation errs in the harmless direction: it is optimistic precisely on the long, heavy-tailed vectors, and those are the vectors whose intact statistic is enormous—no intact sequence with $n \geq 500$ in the whole study has z below 23.6, against a largest null draw of 5.59 over all 2 500 000 draws. Where the margin is thin the approximation is conservative: the weakest intact sequence in the study has $z = 5.7$ at $n = 41$, and in the shortest stratum ($n = 38$) the realised false-alarm rate is *below* nominal at every level we can resolve, with a largest null draw of $z = 4.55$. Second, when a calibrated deep threshold is genuinely required, the Monte-Carlo p -value used for CAT in Section 4.2 applies to AOT unchanged, at the cost of B passes over the vector. Our recommended operating point is justified by this measured separation rather than by a nominal tail probability, and Section 7.2 reports it that way.

What AOT is blind to r is invariant to circular rotation and, up to $O(n^{-1})$ terms, to reversal; it is also insensitive to a permutation that keeps long contiguous runs intact, in the sense that power depends on how

Table 1

Measured tail behaviour of the normal reference. Each row is one sequence; 500 000 uniform permutations were drawn per sequence and scored with the normal rule. *Ratio* is the realised false-alarm rate divided by the nominal level, so > 1 is anti-conservative. The approximation is optimistic on long heavy-tailed vectors, whose intact z exceeds 23.6, and conservative on the short stratum, where the intact margin is thin. Generated by `tools/make_artifacts.py aottail; source artifacts/aot_tail.csv`.

n	kurtosis	10^{-2}	10^{-3}	10^{-4}	10^{-5}	10^{-6}	$\max z$
38	2.1	0.92	0.70	0.30	0.20	0.00	4.55
2545	8.9	1.26	1.77	2.30	2.80	6.00	5.59
2785	10.6	1.21	1.52	2.18	2.60	10.00	5.45
11185	6.7	1.07	1.14	1.34	2.00	2.00	4.91
11425	14.1	1.16	1.47	1.72	1.80	2.00	5.32

much dependence survives. Rotation and reversal are not idle worries—they are what an off-by-one write or a reversed sort produces—so AOT must be read as a diagnostic against near-uniform *scrambling*, not as a proof of alignment. Section 7.2 measures the boundary empirically with block and partial corruptions, and finds it to be tight: the diagnostic covers essentially only the uniform case.

4.2. CAT: the cross-arm agreement test

AOT needs serial dependence. Many settings have none: independent samples, classification, tabular data, or forecasting evaluated on a strided, non-overlapping window grid. In those settings a second structure is sometimes available, because empirical papers rarely evaluate one arm. When K arms (plug-ins, hyperparameter settings, seeds, competing models) are evaluated on the same n samples, their per-sample losses are strongly rank-correlated, because sample difficulty dominates arm identity. If a defective pipeline draws an *independent* permutation for each arm, the agreement collapses. Whether it does draw independent permutations is an empirical question about the pipeline, not a property of the fault, and Section 7.2 shows that in our own archive it usually does not.

Statistic and null Let $E \in \mathbb{R}^{K \times n}$ hold the stored per-sample losses, one row per arm. Define

$$\bar{\rho}(E) = \binom{K}{2}^{-1} \sum_{1 \leq u < v \leq K} \rho_S(E_u, E_v), \quad (8)$$

with ρ_S the Spearman correlation. The null is “each row was independently and uniformly permuted”; we draw B independent row-wise permutations of the observed ranks and report $p = (1 + \#\{b : \bar{\rho}_b \geq \bar{\rho}\}) / (B + 1)$, which is exact for finite B . This Monte-Carlo p -value is the one we use.

The analytic reference, corrected An earlier version of this work also reported a normal reference $z = \bar{\rho}\sqrt{n-1}$. That expression is wrong: it is the scale appropriate to a *single* Spearman correlation, whereas (8) averages $M = \binom{K}{2}$ of them. Under the null the M pairwise correlations are uncorrelated to leading order, so $\text{Var}_{H_0}[\bar{\rho}] \approx 1/[M(n-1)]$ and the correct reference is

$$z = \bar{\rho} \sqrt{M(n-1)}, \quad M = \binom{K}{2}. \quad (9)$$

For the $K = 4$ arms used throughout this study the omitted factor is $\sqrt{6}$, so the earlier statistic understated z by a factor of about 0.4082. We verified (9) against the Monte-Carlo null on the 24 independent-permutation groups: the corrected analytic p -value agrees with the Monte-Carlo p -value to 0.0215 in the worst case and 0.0042 at the median, against 0.2051 and 0.1167 for the earlier form, and the ratio of the corrected theoretical standard deviation to the Monte-Carlo one is 1.0039 at the median. We report (9) for completeness and use the Monte-Carlo p -value in all verdicts, because (9) inherits the independence assumption that Section 7.2 shows to be the binding constraint.

Properties and the assumption that matters CAT requires no serial structure, so it works exactly where AOT does not. It does require $K \geq 2$ arms stored against the same sample set, and it requires those arms to have been permuted *differently*. It is invariant to a permutation applied identically to every arm, and—this is the part that matters in practice and that we underestimated—the invariance degrades gracefully rather than abruptly, so *partial* sharing is nearly as damaging as full sharing. A pipeline that shuffles all arms with one shared seed, or that sorts all arms’ outputs by some common key, or that merely reuses a loader seed across two of four arms, retains enough cross-arm rank structure to pass CAT while being thoroughly misaligned. Conversely a pipeline that scrambles arms independently on non-overlapping windows fails CAT and cannot be judged by AOT. The two diagnostics are complementary in the transformations they cover, which is why we run them as a battery; they are not jointly complete, and Section 7.2 quantifies the gap.

4.3. The battery, and its honest reading

Given a set of stored per-sample vectors we run AOT on each vector and CAT on each group of arms that share a sample set, and we report a per-artefact verdict at a stated α . We recommend $\alpha = 0.05$ for a warning and a far smaller level for a hard build failure: because the intact signal is enormous (Section 7.2 reports a median z of 52.8), a very conservative threshold costs almost nothing in missed certifications—at $\alpha = 10^{-12}$ we still certify 97.5% of intact sequences, and 0% of permuted ones—while the sequences it does fail to certify are the short ones, where the evidence genuinely is thin. The strict threshold must be read as a z margin, not as a nominal error probability: our tail calibration (Table 1) shows the normal rule is anti-conservative in the deep tail on long, heavy-tailed sequences, by a factor of up to 2.8 at 10^{-5} . What justifies a build-breaking threshold here is the measured gap between the two populations—the smallest intact z we observed is 5.7 (and 23.6 once $n \geq 500$) against a largest permuted z of 3.35—not the nominal level attached to it.

Three things the battery is *not*. It is not a proof of alignment: it rejects near-uniform scrambling, not rotation, reversal, block permutation, mild partial permutation, or an order shared between arms (Section 9). It is not a general integrity test: on the injected corruptions of Section 7.2 the covered region of fault space is narrow, and on the real defective groups in our archive CAT is inside its blind spot far more often than outside it. And it is not a substitute for write-side provenance: a test can only be run by someone who thinks to run it, whereas a contract fails closed. We therefore present the battery as a targeted audit tool for artefacts that already exist—including artefacts one has downloaded rather than produced—and the contract of the next section as the primary defence.

5. A provenance contract for per-sample artefacts

The fault is a broken assumption about an array’s index. The fix is to stop leaving that assumption implicit. The database and workflow communities have long insisted that a derived artefact carry a record of how it was produced [22, 23, 24, 25], and the machine-learning tooling stack tracks hyper-parameters, metrics and code versions [26]. What none of them tracks is the one property that a positional join depends on: *which order the rows are in*. We propose to make that explicit, and we deliberately choose the smallest mechanism that fails closed.

The contract Every per-sample artefact is written together with a sidecar record. An earlier version of this work recorded only the order the array claimed to be in, drawn from a closed vocabulary, plus its length. That is not enough, and the gap is worth stating because it is the gap a reader would assume closed: an equally long array from a different horizon, an array rotated by one position, an array permuted arbitrarily, and a sidecar left behind by a previous run all pass a label-and-length check. The contract we now specify and implement binds each array to the identity of the samples it describes.

The record carries: (i) a `schema_version`, so a reader that does not recognise the format refuses rather than guesses; (ii) the `split`; (iii) `n_samples`; (iv) an `ordered_sample_id_digest`, a hash of the *ordered* list of canonical sample identifiers, which pins both the sample set and its order; (v) a `value_digest`, a hash of the array’s own bytes, which pins the array to this sidecar and not to a previous one; (vi) `order_semantics` from a closed vocabulary (`dataset_index`, `loader_emission`); (vii) the cell identity (`dataset`, `backbone`, `plug-in`, `look-back`, `horizon`, `seed`); and (viii) the `producer_commit`, recorded explicitly as `unknown` when it cannot be determined rather than omitted.

The design principle The point that does the work is not the list of fields but the following rule. Every sample must have a stable identifier, and consumers must join *by that identifier*, not by position. For sliding-window forecasting the natural canonical identifier is the window’s origin in the underlying series, qualified by the sample set it belongs to; we use (dataset, split, look-back, horizon, window_origin), which is deterministically reconstructible and therefore back-fillable over an existing archive. Where a positional read is required for performance, the reader must compare ordered sample-identifier digests *before* reading, and raise on mismatch. Readers do not open arrays; they call a loader that takes the *required* order as an argument and validates the whole record, raising a typed exception on a missing field, an unrecognised schema, a digest mismatch, a split mismatch, a length mismatch or an out-of-vocabulary order label. Order-symmetric consumers—anything computing a mean—explicitly pass `None` and thereby document that they do not care. Legacy artefacts written before the contract are accepted only when the caller opts in explicitly, and that path warns; the default is to fail closed.

Listing 3: The read side of the contract. The default is to require index order and full validation, so a positional join cannot be written by accident; an order-symmetric consumer must say so explicitly, and a pre-contract artefact must be opted into explicitly.

```

1 def load_series(run_dir, name, require_order="index",
2               expect_n=None, legacy_ok=False):
3     """Load a per-sample array under the provenance contract."""
4     side = run_dir / f"{name}.provenance.json"
5     if not side.exists():
6         raise MissingFieldError(f"{name}: no provenance sidecar; refusing to load")
7     meta = json.loads(side.read_text())
8     values = np.load(run_dir / f"{name}.npy")
9     if is_legacy_sidecar(meta):
10         if not legacy_ok:
11             raise SchemaVersionError(
12                 f"{name}: sidecar predates {SCHEMA_VERSION}; "
13                 f"pass legacy_ok=True to accept it unverified")
14         warnings.warn(f"{name}: legacy sidecar, identity unverified",
15                     ProvenanceWarning)
16     return values
17     validate_sidecar(meta, values, require_order=require_order, expect_n=expect_n)
18     return values

```

The write side Producing an index-order artefact from a shuffling loader does not require disabling the shuffle—which, as Section 7.6 shows, has a cost. It requires recording *which* sample each entry belongs to. We wrap the dataset so that each item carries its own index, accumulate losses into a preallocated vector at those indices, and write the result as `dataset_index` order; the loader-order vector and the visiting permutation are written alongside, tagged honestly, so that the defect can be reconstructed for study rather than only avoided. The recorder is streaming: it stores one scalar per sample, never the predictions, so the memory cost is $O(n)$ floats and the artefact for our longest split is under 100 kB. Because the canonical identifier is reconstructible, we were able to back-fill the contract over the whole existing archive: 356 cells and 1424 sidecars, with 0 failures, after which every array in the archive reads under the strict path and every analysis in this paper reproduces bit-for-bit.

Cost and coverage The sidecar is a few hundred bytes; the reader adds one file read and two hashes. To characterise what this buys we injected 25 distinct identity faults and recorded which clause fires (Table 2). The contract stops 20 of them, including the transformations the diagnostics of Section 4 cannot see: circular rotation, arbitrary permutation, truncation, a stale sidecar, an equally long array from a different horizon or split, and a loader-order array mislabelled as index order. An adversarial suite of 33 tests exercises these paths, inside a repository suite of 99 tests.

It does not stop 5, and we list them rather than leave them implicit, because the failure mode this paper is about is precisely a safeguard trusted beyond its scope. The most important is this: if two producers independently write artefacts using the *same* wrong loader order, each sidecar is internally consistent and both declare the same ordered identifiers, so the contract sees a legitimate join. Nothing in the record can

Table 2

Machine-generated coverage of the provenance contract over 25 injected identity faults: 20 are stopped by a named clause, 5 are not. The uncovered rows are reported in full because a contract trusted beyond its scope reproduces the failure mode this paper is about. Generated by tools/make_contract_coverage.py; source artifacts/contract_coverage.csv.

injected fault	clause that fires	outcome
single_element_edited	value_digest	DigestMismatchError
float_perturbation_1e-12	value_digest	DigestMismatchError
cyclic_rotation_of_the_whole_array	value_digest	DigestMismatchError
random_permutation_of_the_array	value_digest	DigestMismatchError
array_truncated_by_one_sample	n_samples	IdentityMismatchError
stale_sidecar_array_rerun_in_place	value_digest	DigestMismatchError
array_replaced_by_another_cell_same_length	value_digest	DigestMismatchError
val_product_consumed_as_test	split	IdentityMismatchError
same_length_different_horizon_joined	ordered_sample_id_digest	IdentityMismatchError
same_length_different_dataset_joined	ordered_sample_id_digest	IdentityMismatchError
loader_order_relabelled_as_index_order	order_semantics+ordered_sample_id_digest	IdentityMismatchError
sample_id_source_replaced_val_perm	ordered_sample_id_digest	IdentityMismatchError
positional_join_across_different_sample_sets	ordered_sample_id_digest	IdentityMismatchError
join_of_a_product_with_no_sample_ids	sample_id_spec	MissingFieldError
legacy_sidecar_without_schema_version	schema_version	SchemaVersionError
unknown_future_schema_version	schema_version	SchemaVersionError
required_field_removed_from_sidecar	ordered_sample_id_digest	MissingFieldError
order_semantics_outside_the_enumeration	order_semantics	OrderSemanticsError
sidecar_file_deleted	schema_version	MissingFieldError
duplicate_sample_ids_in_one_product	ordered_sample_id_digest	IdentityMismatchError
<i>faults the contract accepts without error (blind spots)</i>		
two_producers_share_one_wrong_loader_order	(none)	accepted
<i>each product is self-consistent and both declare the same ordered ids, so the contract sees a legitimate loader-order join; only the AOT/CAT detectors or the raw dataset indices can reveal that the order is semantically wrong</i>		
no_sidecar_but_consumer_passes_require_order_	(none)	accepted
None		
<i>kept for order-invariant consumers of pre-contract arrays; warns loudly but does not raise, so it is an explicit opt-out hole rather than a checked path</i>		
positional_join_truncated_to_n_min	(none)	accepted
<i>when products have different lengths the ordered digests are not comparable, so only dataset/split/seq_len/pred_len are verified and a prefix misalignment survives</i>		
upstream_data_or_preprocessing_changed	(none)	accepted
<i>sample ids are (dataset, split, seq_len, pred_len, window_origin) and carry no hash of the raw series, so a re-derived product from changed input data validates and joins cleanly with features computed from the other version</i>		
per_sample_value_computed_wrongly_but_consistently	(none)	accepted
<i>the contract certifies which sample a number belongs to, never that the number is the right statistic; a wrong reduction with a fresh digest is indistinguishable</i>		

reveal that the shared order is semantically wrong; only a diagnostic such as AOT, or the raw dataset indices, can. The remaining four are: an explicit opt-out for order-invariant consumers of pre-contract arrays, which warns but does not raise; a positional join truncated to the shorter of two lengths, where ordered digests are not comparable and only cell identity is checked; a change in the upstream data or preprocessing, since the identifier carries no hash of the raw series; and a per-sample value that is computed wrongly but recorded consistently, since the contract certifies which sample a number belongs to and never that the number is the right statistic. The contract's other structural weakness is the mirror image of its strength: it protects only artefacts written by code that adopts it, which is why the diagnostics of Section 4 remain necessary for everything already published. Section 8 states how we think the two should be combined.

6. Study design

The theory of Sections 2–3 is exact and needs no experiment. What needs an experiment is everything quantitative: how much serial structure real per-sample error vectors actually have, whether the detectors'

Table 3

The grid. Every cell is one training run; *arms* counts the plain backbone plus the plug-ins. Generated from `artifacts/runs.csv`.

backbone	datasets	horizons	arms	runs	GPU hours
DLinear	7	8	4	272	15.1
PatchTST	7	4	3	42	3.1
iTransformer	7	4	3	42	3.0
all	7	8	4	356	21.3

nominal error rates hold on real data rather than on synthetic draws, how severe a corruption has to be before it is detectable, how much a real downstream conclusion moves, and what the repair costs. We therefore run a grid whose purpose is to produce *artefacts*, not a leaderboard.

6.1. Grid

We use 7 standard public forecasting benchmarks (ETTh1, ETTh2, ETTm1, ETTm2, Weather, Exchange-Rate and ILI), 3 backbones spanning the three dominant architectural families—a linear model [8], a patch-based Transformer [9] and an inverted-attention Transformer [10]—and 4 arms: the plain backbone and three model-agnostic plug-ins, reversible instance normalisation [27], a frequency-domain loss [28] and a statistics-prediction normaliser in the style of Liu et al. [29]. Horizons are {96, 192, 336, 720} and {24, 36, 48, 60} for ILI, with a look-back of 96 (36 for ILI). Training is the standard protocol of the reference library [4]: Adam at 10^{-4} , mean squared error, at most 10 epochs, early stopping with patience 3, batch size 32, and the library’s own splits. Multivariate-to-multivariate forecasting throughout. All 356 runs completed on a single NVIDIA V100 (32 GB) in 21.3 GPU hours and 2 945 trained epochs; Table 3 breaks the budget down.

The grid has two stages with different jobs.

Stage E1 maximises the diversity of per-sample sequences available for calibrating the detectors: all datasets, horizons and arms for the linear backbone, and a reduced factorial for the two Transformers. Every E1 run uses a deterministic evaluation loader, so its stored vector is a ground-truth index-order sequence. This is what lets us measure a false-alarm rate on *real* data: we corrupt a known-good sequence ourselves and ask the detector what it says, rather than assuming a synthetic AR(1) model of per-sample loss.

Stage E2 produces matched pairs. For four datasets, two horizons, four arms and three seeds we run each configuration twice, changing exactly one flag—the validation loader’s shuffle—and nothing else. The shuffled member of each pair writes both the loader-order vector and, via the recorder, the index-order vector, which gives us the *observed* defect rather than a simulated one. The pairs also answer the maintenance question of Section 7.6: what does the one-line repair do to the reported numbers?

6.2. What is recorded

For every run we store: the per-sample loss vector for validation and test in index order; for shuffled runs, additionally the loader-order vector and the visiting permutation; the aggregate metrics; the selected epoch; the wall-clock time; and the resolved configuration. Every array carries the sidecar of Section 5, and every analysis in this paper loads arrays through the contract, so a mis-ordered array cannot enter our own results—an instance of eating one’s own cooking that we consider part of the contribution. Per-sample features are computed from the look-back window only, never from the target, so they are legitimate covariates for a difficulty analysis.

6.3. Analyses

Two downstream analyses stand in for the class of per-sample claims. **T1 (difficulty association)** asks the descriptive question a paper prints in a table: does per-sample loss correlate with cheap per-sample covariates, and how much harder is the top covariate decile than the bottom? It also asks the predictive version—fit a gradient-boosted model on the joined validation table and measure rank agreement on the untouched test split—because that is what a paper claims when it says difficulty is predictable. We report rank agreement rather than R^2 because validation and test cover different calendar periods, so their error

levels differ by construction; and we use a purged holdout [18] inside the validation split to avoid crediting a model for window overlap. **T2 (per-sample arm selection)** asks the decision question: train a gate on the joined validation table to pick one of the 4 arms per sample, and measure the realised test error against the best fixed arm and against the per-sample oracle. Each analysis is run under three conditions—*index* (correct join), *defect* (join against the loader-order vector), and *control* (correct vector, permuted feature rows)—which is exactly the triple that Proposition 5 says must collapse to two.

6.4. Reproducibility

The manuscript contains no typed number. Each quantitative claim expands a LaTeX macro generated from the artefacts, and a verifier recomputes every macro from the artefacts and fails on any mismatch, reporting the macro name, the printed value and the recomputed value. Tables and figures are generated by the same script. The test suite that accompanies the release covers the propositions as executable statements—aggregate invariance, length blindness, distributional equality of defect and control, the randomisation moments against brute-force enumeration of all $n!$ permutations for $n \leq 8$, the false certification rate at a nominal level—so a reader can check the claims of Sections 2–4 without a GPU.

One detail of the analysis layer is worth reporting, because it is an instance of the class of fault this paper is about and we did not get it right the first time. Every simulated permutation in the study—the corrupted arm of each detector comparison, the feature shuffle of each control, the sub-sampling of the battery—is drawn from a generator seeded by a content-derived key. An earlier version of our own analysis code seeded those generators with Python’s builtin `hash()` of the cell identifier. That function is salted per interpreter for strings, so the artefacts were not reproducible across processes: re-running the analysis produced different permutations and therefore different numbers, while every internal consistency check still passed, because the verifier and the artefacts moved together. The fault was silent in exactly the sense of Section 2.3. Seeds are now derived with a fixed cryptographic hash (`src/seeds.py`), a regression test asserts that seeding agrees across interpreters started with different hash salts, and a second test greps the analysis layer for a relapse. Re-running `tools/make_artifacts.py` all now reproduces the shipped artefacts, and hence every number in this paper, bit for bit.

7. Results

7.1. The permutation is real, and the aggregate does not move

We first confirm on real artefacts what Section 2.3 proves. In the 96 runs whose validation loader shuffles, we hold both the loader-order vector the pipeline would have saved and the index-order vector the recorder reconstructed. In 100% of them the two are exact rearrangements of each other, in 100% they have identical length, and the largest relative difference between their means over all 96 runs is 0, i.e. at most the last bit of a double-precision sum. Every order-symmetric aggregate a reader would see is therefore identical, while the vector a per-sample analysis consumes is a uniformly random rearrangement. Running AOT on those 96 *observed* loader-order vectors—the real defect, not a simulation of it—gives a median serial correlation of -0.0003 and certifies 9.38% of them, matching the behaviour of the simulated permutations we use for the calibration below.

Figure 1 shows what the two vectors look like. The intact sequence is visibly a time series with long excursions; the same values in loader order are indistinguishable from noise; the lag-1 scatter separates the two completely. This is the entire basis of the detector, and it is worth noting how much information the aggregate throws away: the two panels have the same mean, the same variance, the same histogram, and nothing else in common.

7.2. Detector operating characteristics

Across 712 real index-order sequences (lengths from 38 to 11 425), together 4 265 096 window-level evaluations, the circular serial correlation has median 0.9975 and minimum 0.85: window overlap produces dependence close to the maximum possible, uniformly over datasets, horizons, backbones and arms. That count is a sum over sequences, not a count of distinct windows: the sequences are drawn from 56 distinct evaluation sample sets holding 287 496 windows in total, and each is audited once per run and arm, so sequences that share a dataset, horizon and split are not independent replicates. Section 9 returns to what

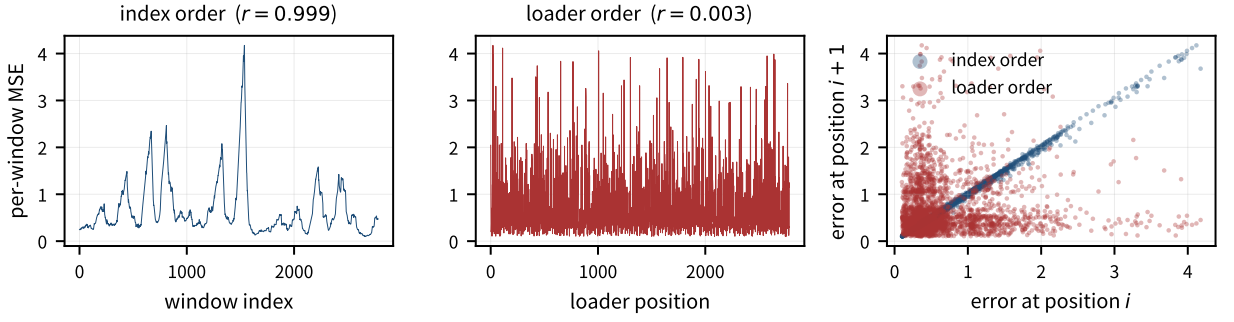


Figure 1: One real validation error vector in index order (left) and the same values in loader order (middle), with the lag-1 scatter of both (right). The two vectors have identical means, quantiles and histograms.

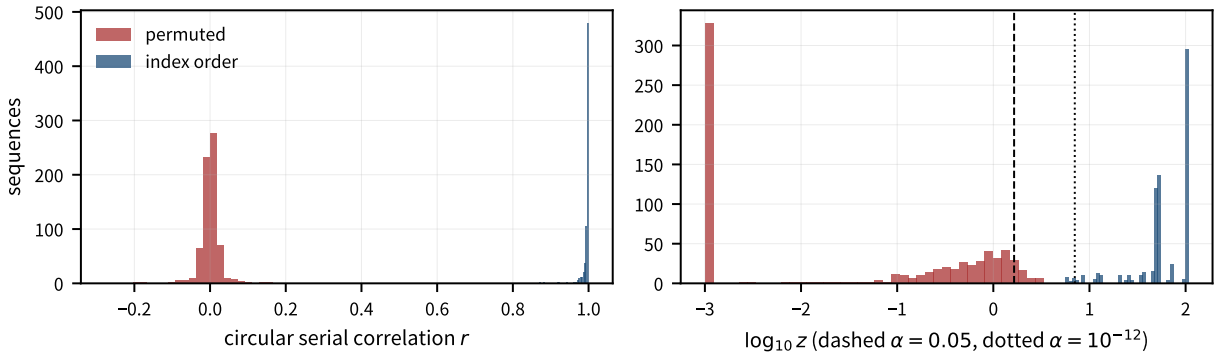


Figure 2: Separation of the AOT statistic on 712 real sequences and their permutations. Left: the raw statistic. Right: the standardised statistic on a log scale, with the $\alpha = 0.05$ and $\alpha = 10^{-12}$ thresholds.

that does and does not license. After a uniform permutation of the same sequences the median falls to 0.0011 with maximum 0.4617. The standardised statistic separates the two populations by orders of magnitude (Figure 2): median $z = 52.8$ for intact sequences with a minimum of 5.7, against a maximum of 3.35 over all permuted sequences.

At $\alpha = 0.05$ the test certifies 100% of intact sequences and 6.18% of permuted ones. The latter is the realised false certification rate on vectors drawn exactly from the null; it sits slightly above the nominal 5%, consistent with the mild anti-conservatism of the normal reference measured in Table 1. Table 4 sweeps the level, and its deep rows should be read as thresholds on z rather than as calibrated error probabilities. The row we recommend in practice is $\alpha = 10^{-6}$, i.e. $z > 4.75$: it separates the two populations *perfectly* on our data, certifying 100% of intact sequences and 0% of permuted ones, and the separation—not the nominal level—is what supports it. The margin is large: $z_{\min} = 5.7$ on intact sequences against $z_{\max} = 3.35$ on 712 permuted ones, and 5.59 over the further 2 500 000 null draws of Table 1. Pushing to 10^{-12} costs sensitivity only on the shortest artefacts: 97.5% of intact sequences are still certified, and every one of the 18 that are not has at most 50 windows—AOT’s evidence grows with n , and an evaluation split of a few dozen windows simply does not contain that much of it. A continuous-integration job can therefore fail the build on order corruption at a false-alarm rate that we can bound empirically rather than nominally, provided the artefact is not shorter than a few hundred windows; for shorter artefacts we recommend the Monte-Carlo p -value, whose level is exact by construction.

At the level a practitioner actually uses, the normal approximation is not doing hidden work. On 120 permuted sequences, where the null is true by construction, the analytic and Monte-Carlo p -values differ by at most 0.0262, and the empirical rejection rates at $\alpha = 0.05$ are 9.17% (analytic) and 7.5% (Monte-Carlo).

Table 4

AOT certification rates as a function of the level, over 712 real sequences and their uniform permutations. Generated from `artifacts/certification.csv`.

α	index-order certified (%)	permuted certified (%)
0.05	100.0	6.18
0.01	100.0	1.69
10^{-3}	100.0	0.14
10^{-6}	100.0	0.00
10^{-12}	97.5	0.00

Table 5

Where AOT stops working. Each row is one injected corruption family; *flagged* is the fraction of corrupted sequences that AOT declines to certify, i.e. the detection rate. Block permutations and mild partial permutations retain enough serial dependence that the complete-randomisation null is still rejected, so they are certified and the fault is not flagged. Generated from `artifacts/aot_sensitivity.csv`.

corruption	parameter	order structure kept (%)	sequences	AOT flagged (%)
block-order shuffle, block size b	8	87.5	60	0.00
block-order shuffle, block size b	32	96.9	60	0.00
block-order shuffle, block size b	128	99.2	58	0.00
block-order shuffle, block size b	512	99.8	57	0.00
partial shuffle, fraction f moved	0.05	95.0	60	0.00
partial shuffle, fraction f moved	0.10	90.0	60	0.00
partial shuffle, fraction f moved	0.25	75.0	60	1.67
partial shuffle, fraction f moved	0.50	50.0	60	1.67
partial shuffle, fraction f moved	1.00	0.0	60	95.00

Deeper in the tail it does do work, and Table 1 quantifies exactly how much; the two measurements together are the reason we report a recommended threshold with a measured margin instead of a nominal guarantee.

Milder corruptions, and the boundary of the diagnostic. A uniform permutation is the worst case; real defects can be gentler. Shuffling *batch order* while keeping batches internally sequential—what a batch-level sampler produces—preserves within-block dependence, and permuting only a fraction of positions preserves most of it. This is where AOT stops working, and the boundary is sharper than we expected when we designed the test.

The direction of the statistic has to be kept straight here, because it is easy to read the table backwards. AOT rejects the hypothesis of complete randomisation. A block-permuted vector still has almost all of its serial dependence, so AOT rejects that hypothesis, so the vector is *certified in structured order*—which, on a vector we injected a fault into, means the diagnostic did not flag the fault. Table 5 therefore reports the flag rate, i.e. the fraction of corrupted sequences that AOT declines to certify, and the result is close to zero over most of the fault family. Over the 235 sequence-by-block-length cells the flag rate is 0%: no block permutation at any of the four block lengths we injected was flagged, at any sequence length. Partial permutations of 5% and 10% of positions are likewise flagged in 0% of cases, and moving a quarter or a half of all positions still only reaches 1.67% and 1.67%. Only the fully random permutation is reliably flagged, at 95%.

We draw the conclusion that follows rather than the one we would have preferred. AOT was effective against complete random permutations in our injected faults, and it was not observed to flag the block permutations or the mild partial permutations we evaluated. It should be used as a targeted diagnostic for near-random order corruption—which is exactly the fault the library default of Section 2 produces, and which is why it works on our real artefacts—and not as a general order-integrity test. A local or block-wise statistic would plausibly cover part of the gap; we have not built one, and we prefer to state the limitation than to gesture at a fix we did not evaluate.

Cross-arm agreement. On the 96 groups of arms evaluated against a common sample set, CAT gives median $\bar{\rho} = 0.9371$, certifying 99% of groups; after *independent* per-arm permutation the median falls to -0.002 and 4.17% are certified. 1 intact group is not certified, and it is one of the shortest in the study ($n \leq 41$ windows, against a minimum of 38 over all groups), where four arms simply do not supply enough rank information; its $\bar{\rho}$ of -0.192 is the minimum over intact groups. The agreement is not driven by one dominant pair: across all 492 individual arm pairs the median Spearman correlation is 0.9507. Sample difficulty dominating arm identity is a robust empirical regularity, and CAT converts it into a test—under the independence assumption, which the next subsection examines.

7.3. Whether CAT’s independence assumption holds in a real pipeline

The CAT result just quoted is conditional on each arm having been permuted independently. We stated that assumption when we designed the test and justified it by an argument about process isolation: each arm is a separate process with its own loader and its own random state. That argument is wrong for our own pipeline, and we discovered it only by checking. Because Stage E2 stores the visiting permutation alongside every defective vector, we can audit the assumption directly instead of simulating it.

Across the 24 real defective four-arm groups in the archive, every one of the 96 stored permutations is non-identity, and all 24 groups satisfy the consistency check that the stored loader-order vector equals the index-order vector composed with the stored permutation—so the recorded defect is exactly the defect. The permutations are not, however, independent across arms. Only 2 groups have four pairwise distinct orders; 20 groups share at least one order between two arms, and 2 groups use a single order for all four arms. Of the 144 arm pairs, 11 groups have three identical pairs and 2 have all six. The mean rank correlation between two arms’ permutations has median 0.4874 and ranges from 0.0001 to 1. The mechanism is mundane: arms that share a data configuration draw their loader seed from the same derived key, so the shuffle is reproduced rather than redrawn.

The consequence for CAT is severe. Run on the real loader-order vectors, CAT certifies 22 of 24 groups (91.67%) as retaining cross-arm structure—that is, it declares clean a set of groups every one of which is corrupted. For reference, on the intact vectors of the same groups it certifies 24 of 24, so the diagnostic is not merely noisy: it is answering a different question from the one a user would think they asked. AOT on the same vectors behaves much better, flagging 87 of 96 arms (90.63%) and missing at least one arm in 6 groups, with a smallest observed p -value of 0.00412; the shared-order structure that defeats CAT is irrelevant to AOT, which reads each vector on its own.

Table 6 reports the audit, and Table 7 isolates the mechanism under controlled sharing. Taking the same 24 groups and injecting permutations ourselves, CAT flags every group when the four permutations are drawn independently (0 of 24 certified, median $\bar{\rho} = -0.0023$) and flags none of them under any sharing pattern we tried: all four arms sharing one order (median $\bar{\rho} = 0.9509$), three of four sharing (0.4827), two of four sharing (0.1579), or two disjoint pairs (0.3199). A single shared pair among four arms is enough. The residual agreement it leaves is smaller than the intact value of 0.9509, but the Monte-Carlo null is a null of complete independence, so any residual agreement rejects it.

We therefore restate CAT’s scope. CAT is informative when independently produced artefacts have sufficiently distinct order perturbations. Shared or correlated loader orders preserve cross-arm rank structure and constitute a blind spot that, in the one real pipeline we were able to audit, was the common case rather than the exception. A user of CAT should first establish that the arms’ orders were drawn independently—which, if the visiting permutations are recorded, is a direct check, and if they are not, is an assumption that cannot be verified after the fact. This is a further argument for the write-side contract rather than for the diagnostic.

7.4. Where each test stops working

AOT’s power comes from window overlap, so we remove the overlap deliberately. Sub-sampling every s -th window reduces the shared fraction of two consecutive retained windows from $1 - 1/(L + H)$ to $\max(0, 1 - s/(L + H))$, and at large enough s the retained windows are disjoint. Table 8 and Figure 3 trace both tests along that axis. AOT certifies 100% of intact groups at stride 1 and degrades as overlap falls; on the 16 configurations in which the retained windows are exactly disjoint ($s = 192$, mean length 60) it certifies 2.1%. CAT, which never used serial structure, certifies 98.9% at stride 1 and 100% on the disjoint

Table 6

Audit of the independence assumption behind CAT on 24 real defective four-arm groups. The stored visiting permutations are shared across arms far more often than not, and CAT consequently certifies 91.67% of genuinely corrupted groups as clean. Generated by `tools/make_cat_revision.py`; source `artifacts/observed_perm_scan.csv`.

quantity	value
groups of arms over a common sample set	24
arms (group $\times$ plugin)	96
arms whose validation order is not the identity	96
groups with <code>val_loader_order == val_index_order[val_perm]</code>	24
sharing: one permutation for all arms (<code>all_identical</code>)	2
sharing: some arms share (<code>partial_shared</code>)	20
sharing: every arm distinct (<code>all_distinct</code>)	2
identical arm pairs per group: 0, 1, 2, 3, 6 pairs	2 / 7 / 2 / 11 / 2
mean Spearman between arm permutations, median (min, max)	0.4874 (0.0001, 1.0000)
CAT certifies the observed loader order	22 / 24 (91.67%)
CAT median $\bar{\rho}$, observed loader order	0.4189
CAT certifies the intact index order	24 / 24 (100.00%)
CAT median $\bar{\rho}$, intact index order	0.9509
AOT flags the observed loader order	87 / 96 (90.63%)
arms AOT fails to flag	9
groups with at least one unflagged arm	6 / 24
smallest AOT p over all arms	0.00412

Table 7

CAT under controlled cross-arm order sharing, 24 groups per condition. The diagnostic works only when all four permutations are drawn independently; a single shared pair among four arms defeats it. Generated by `tools/make_cat_revision.py`; source `artifacts/cat_conditions.csv`.

condition	sharing pattern	arm labels	distinct perms.	certified	median $\bar{\rho}$	median p (MC)
intact	intact (no permutation injected)	—	0	24 / 24	0.9509	0.0005
independent	independent permutation per arm	0 1 2 3	4	0 / 24	-0.0023	0.6207
all_shared	one permutation shared by all four arms	0 0 0 0	1	24 / 24	0.9509	0.0005
shared_k3	three arms share, one independent	0 0 0 1	2	24 / 24	0.4827	0.0005
shared_k2	two arms share, two independent	0 0 1 2	3	24 / 24	0.1579	0.0005
two_pairs	two disjoint pairs, each sharing	0 0 1 1	2	24 / 24	0.3199	0.0005

grid. Corrupted inputs, pooled over all 758 configurations of the sweep, are certified at 6% by AOT and 3.7% by CAT—the nominal level, up to the sampling noise of 95 groups per stride—so the loss of power at long strides is a loss of *sensitivity*, not a gain in false alarms. The practical rule follows directly: use AOT when samples overlap, use CAT whenever two or more arms share a sample set, and treat an artefact that satisfies neither precondition—a single arm on independent samples—as unauditable after the fact, which makes the write-side contract mandatory rather than advisable.

7.5. Downstream: a real conclusion changes, and the control does not

T1. Under the correct join, per-sample loss is genuinely associated with cheap look-back covariates: across 356 runs the median Spearman correlation between look-back volatility and loss is 0.2192 (mean 0.1668), the best single covariate reaches 0.2817 in absolute value, and the top volatility decile is $1.426\times$ harder than the bottom decile. Under the defect the same numbers become -0.0004 , 0.0267 and $0.995\times$: the effect is gone, the decile contrast is exactly the ratio of two samples from the same distribution, and a reader would conclude that per-sample difficulty is unrelated to the obvious drivers of per-sample difficulty. The predictive version tells the same story—median transfer rank agreement on the untouched test split falls from 0.2275 to 0.0156, and the purged within-split holdout from 0.1191 to 0. Table 11 and Figure 4 collect the comparison.

Table 8

The battery against loss of overlap. *Overlap* is the mean shared fraction of consecutive retained windows; *clean* and *perm.* are certification rates (%) on intact and permuted vectors; the permuted columns estimate the nominal 5% level from 95 groups per stride, so their per-row noise is large while the pooled rate quoted in the body is not. Generated from `artifacts/battery.csv`.

stride	overlap	n	AOT clean	AOT perm.	CAT clean	CAT perm.
1	1.00	5740	100.0	5.5	98.9	9.5
4	0.99	1566	100.0	4.7	100.0	1.1
16	0.94	410	100.0	6.9	100.0	2.4
64	0.77	108	83.7	6.6	100.0	1.3
192	0.31	59	28.7	6.5	100.0	2.8

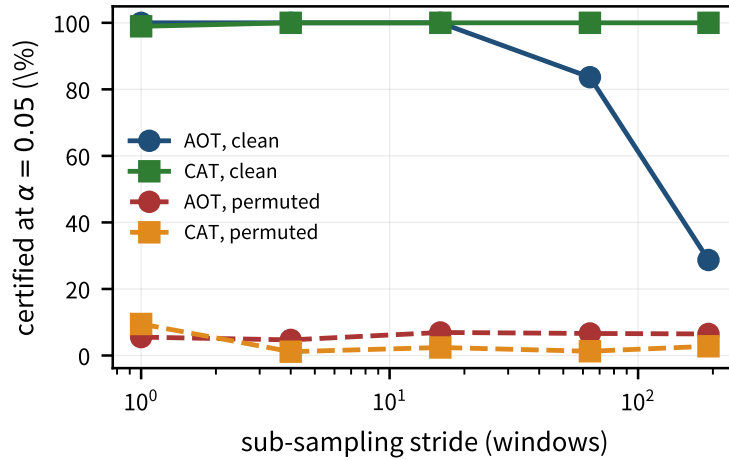


Figure 3: Certification rate against sub-sampling stride. AOT follows the overlap; CAT does not depend on it. Permuted input is certified only at the nominal level, at every stride and by both tests.

How significant is that association, really? An earlier version of this work reported that the intact association is significant at $\alpha = 0.05$ in 89.33% of runs, using the standard independent-pairs approximation $p = 2\Phi(-|\rho|\sqrt{n-1})$. That figure is not defensible and we withdraw it. The windows are sliding with stride one, so consecutive per-window errors have the lag-1 correlation of at least 0.85 that this paper’s own detector relies on; treating n overlapping windows as n independent pairs under-states the standard error by a large factor, and it does so in the anti-conservative direction. This is the same category of error as the one the paper is about—a statistic that reads a structured artefact as an exchangeable collection—so we correct it in kind rather than caveat it.

We replace the figure with two dependence-aware readings, both on the same 356 runs (Table 9). Under a circular block-permutation null with block length $\max(L + H, \lceil n^{1/3} \rceil)$, chosen so that a block spans the entire window-overlap range, the association is significant in 34.27% of runs. Under a moving-block bootstrap with the same block length, the 95% interval for ρ excludes zero in 32.87% of runs. The two agree closely, and both are roughly a third of the naive figure. The block-permutation null is well calibrated where we can check it: on the defect and control conditions, where the true association is zero by construction, its rejection rate is close to the nominal 5%.

The descriptive effect itself is unchanged and remains the substantive result: median $\rho = 0.2192$ with interquartile range $[0.07, 0.2925]$ over the 356 runs. What changes is how much generality that median supports. The 712 sequences in the study are not 712 independent experiments: they come from 56 distinct evaluation sample sets (28 on the validation side) over 7 datasets, so the effective number of independent units is smaller by more than an order of magnitude. Resampling clusters rather than rows, the median ρ has

Table 9

The intact difficulty association under three inference schemes, over 356 runs. The independent-pairs approximation, which an earlier version of this work used, over-states significance roughly threefold relative to dependence-aware alternatives; cluster resampling over 28 sample sets or 7 datasets widens every interval substantially. Generated by `tools/make_downstream_revision.py`; sources `artifacts/t1_difficulty_depaware.csv` and `artifacts/t1_cluster_summary.csv`.

statistic over cells	point	95% CI, sample-set clusters	95% CI, dataset clusters
median ρ (volatility, per-window error)	0.2192	[0.1049, 0.2731]	[0.0871, 0.2812]
significant at $\alpha = 0.05$, iid approximation (%)	89.33	[79.88, 95.64]	[73.19, 97.98]
significant at $\alpha = 0.05$, block permutation (%)	34.27	[10.05, 57.93]	[8.43, 60.67]
moving-block bootstrap interval excludes 0 (%)	32.87	[11.14, 55.15]	[9.16, 53.09]
median ρ , defect – control	−0.000526	[−0.002811, 0.000868]	[−0.001907, 0.000465]
clusters resampled	356 cells	28 sample sets	7 datasets

a 95% interval of [0.1049, 0.2731] with sample sets as clusters and [0.0871, 0.2812] with datasets as clusters. The association is positive under every reading, but its magnitude is not pinned down to better than a factor of two across datasets, and we make no claim that it would hold at this strength on data outside this grid.

The one conclusion that does *not* weaken under cluster resampling is the central one. The paired difference between the defect and the control has median −0.000526, with a cluster interval of [−0.002811, 0.000868] over sample sets and [−0.001907, 0.000465] over datasets. Both contain zero and both are two orders of magnitude narrower than the effect they would have to contain to distinguish the two conditions. Control degeneration survives the conservative reading, which is what Proposition 5 predicts: it is a distributional identity, not an empirical regularity that could have been an artefact of the grid.

The purged holdout was not purged enough. The predictive arm of T1 fits a difficulty model inside the validation split and evaluates it on a held-out tail of the same split, with a purge between the two so that overlapping windows cannot leak. An earlier version of this work used a fixed purge of 200 window origins for every configuration. That is a constant where the quantity it has to dominate is not: a single window covers $L + H$ time steps, and at $L = 96$, $H = 720$ that is 816. With a purge of 200 the origins on either side of the boundary differ by 401, so the last training window and the first evaluation window still share 415 time steps, of which 319 are shared *prediction targets*. The holdout was therefore partly in-sample at long horizons, and 28 cells in the grid leak this way.

We replace the constant with two rules derived from the window geometry, and we report the realised boundary rather than the intended one (Table 10). Under the `labels` rule, origins must differ by at least H , so no training label is available later than the first evaluation decision; at $L = 96$, $H = 720$ this means a purge of 360 origins, a realised gap of 721, and 0 shared target points—the 95 time steps that still overlap lie entirely inside the evaluation window’s look-back, which is a legitimate covariate. Under the stricter `disjoint` rule, origins must differ by at least $L + H$, giving a purge of 408 and 0 shared time steps of any kind. Both rules are verified by interval arithmetic on the actual window extents rather than by a constant, and the verifier asserts the realised boundary for every cell.

The cost of doing this correctly is that some configurations no longer support the analysis at all. Requiring at least 200 windows on each side of the purge, 310 of 356 cells remain eligible under either rule and 46 do not; the excluded cells are the short-series configurations, all four ILI horizons and the three longest Exchange horizons, where the validation split holds only a few dozen to a few hundred windows. We exclude them and say so, rather than shrinking the purge until the arithmetic fits.

The third column of Table 11 is the point of the paper. The shuffled-feature control—the check an author would run to convince a reviewer that the null in column two is real—lands on the same values as the defect. We certify this rather than assert it. A two one-sided test of equivalence [20, 21] on the paired difference between the defect and the control, with an equivalence margin of 0.03 Spearman units (a fifth of the intact effect), gives a mean difference of 0.00055 with a 95% confidence interval of [−0.00495, 0.00604] and $p = 4.46 \times 10^{-23}$: on the descriptive statistic the two conditions are statistically equivalent, not merely indistinguishable. Because cells that share a dataset share a sample set and are therefore not independent,

Table 10

Purge audit at $L = 96$, $H = 720$, where a window covers 816 time steps. The fixed purge used in an earlier version of this work leaves 319 shared prediction targets across the holdout boundary. Boundaries are verified from window extents, not assumed. Generated by `tools/make_downstream_revision.py`; source `artifacts/t1_purge_audit.csv`.

purge rule	purged origins	required gap	actual gap	shared time pts.	shared target pts.	leaking cells	eligible cells
legacy, fixed purge	200	—	401	415	319	28	—
labels (targets disjoint)	360	720	721	95	0	0	310
disjoint (inputs and targets disjoint)	408	816	817	0	0	0	310

Table 11

T1 under the three conditions, medians over 356 runs. The *defect* and *control* columns are two samples from the same distribution (Proposition 5); the test in the body certifies that they are, rather than merely failing to distinguish them. Generated from `artifacts/t1_difficulty.csv`.

statistic (median over runs)	index	order	defect	control
$\rho(\text{volatility, error})$	0.219	-0.000	0.001	
ρ of the best feature	0.268	0.010	0.009	
top/bottom decile error ratio	1.43	1.00	1.00	
transfer rank agreement on test	0.227	0.016	-0.008	
significant association at $\alpha = 0.05$ (%)	89.3	3.9	5.3	

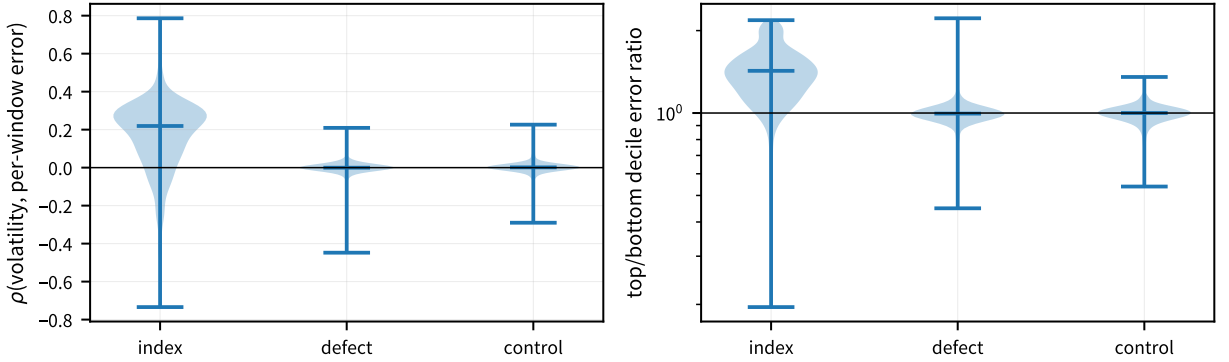


Figure 4: T1 across 356 runs under the correct join, the defect and the shuffled-feature control. Left: association between look-back volatility and per-sample loss. Right: hardness ratio between the top and bottom volatility decile (log scale; 1 is no effect).

we repeat the test at the level of the 7 datasets, averaging the paired differences within a dataset first; the descriptive verdict survives that conservative reading, with interval $[-0.00286, 0.00495]$ and $p = 9.00 \times 10^{-7}$.

The predictive statistic needs a more careful statement, and we prefer to give it than to round it away. On the 324 runs where transfer rank agreement is defined, the run reported in Table 11—which uses the loader’s own recorded permutation wherever we observed one—gives a paired difference of 0.01595 with interval $[-0.001, 0.0329]$ against a margin of 0.03 and $p = 0.0517$: the interval brushes the margin and equivalence is *not* certified at the 5% level for this statistic in this draw. The reason is Monte Carlo, not structural. Both the defect and the control are random draws, so a single-draw equivalence test is itself a random experiment, and the predictive statistic—which requires fitting a model per condition—carries far more of that variance than the model-free descriptive one. Repeating the whole comparison over 8 independent draws of both conditions makes this explicit: every draw certifies equivalence, the largest single-draw p is 0.0215 and the median is 0.0125, the largest single-draw absolute difference is 0.0125—inside the margin—and the draw-averaged difference is 0.004 with interval $[-0.0022, 0.0102]$ and $p = 2.24 \times 10^{-15}$. The same replication on the descriptive statistic gives a draw-averaged difference of 0.0001 with interval $[-0.0023, 0.0024]$. The

Table 12

T2 under the three conditions, medians over 96 groups. Unlike T1, the *defect* and *control* columns are not two samples from the same distribution: the label here is derived from all arms, so Proposition 7 applies and the control is the milder condition. Generated from `artifacts/t2_selection.csv`.

statistic (median over blocks)	index	order	defect	control
gain over best fixed arm (%)	-1.03	-1.12	-0.29	
per-window oracle headroom (%)	8.10	8.10	8.10	
agreement with the per-window oracle	0.330	0.295	0.333	
blocks with a positive gain (%)	25.0	8.3	15.6	

honest summary is therefore: the descriptive reading of T1 certifies equivalence under every reading we tried, including the dataset-clustered one; the predictive reading certifies it in the replicated experiment and sits on the decision boundary in a single draw, and its dataset-level test, with only 7 independent datasets, is too underpowered to certify anything ($[0.0029, 0.0378]$, $p = 0.106$). None of this is evidence that the defect and the control differ; Proposition 5 says they cannot, for statistics of this kind, and what these tests add is a measured bound on how far apart a finite experiment can place them.

T2. We stress at the outset what T2 is and is not. It is a probe of how the fault and its control behave on a derived-label analysis; it is *not* evidence that per-window gating works. On our grid it does not work even with a correct join: over 96 groups a gate trained on the correctly joined table changes test error by a median of -1.032% relative to the best fixed arm—a negative number, i.e. the gate is *worse* than simply keeping the best arm—against a per-window oracle headroom of 8.1% . A positive gain occurs in only 25% of groups, and the Wilcoxon signed-rank test against zero, $p = 8.83 \times 10^{-9}$, is significant *in the negative direction*: the honest summary of the correct join is a reliable small loss, which is consistent with the prior literature reporting that per-sample routing of this kind is not learnable. Under the defect the median gain is -1.115% and under the control -0.287% ; all three conditions are non-results in the sense of Proposition 4, which is exactly why the comparison between them, rather than their absolute level, is what carries information here.

Here, however, the defect and the control are *not* interchangeable, and Proposition 7 says why. The paired defect-minus-control difference in gain is -0.587 percentage points with interval $[-1.009, -0.166]$, which excludes zero as well as the equivalence margin of 0.25 : the defect is measurably *worse* than the control, so here the equivalence test correctly declines to certify equivalence (the dataset-clustered interval is $[-1.279, 0.035]$, $p = 0.892$). The label law is where the difference lives. Over the 96 groups, the per-window best-arm label under the correct join puts on average 46.1% of its mass on its modal arm, with mean entropy 1.653 bits out of a possible 2 ; under independent per-arm permutation the modal mass falls to 33.3% and the entropy rises to 1.822 bits, and the share of windows won by the globally best fixed arm falls from 45.7% to 33.1% . The defect thus flattens the label towards uniform, which is exactly the product-law statement of Proposition 7. A gate trained on such labels distributes windows across arms nearly independently of arm quality, and the realised error drifts towards the uniform-arm baseline, which on this grid costs 3.99% relative to the best fixed arm. The control, which keeps the label intact and destroys only the features, degenerates to predicting the modal arm and therefore costs almost nothing. The reading is uncomfortable and worth stating plainly: for derived-label analyses the standard negative control is not merely powerless, it understates the damage. We draw a deliberately limited conclusion from T2—it does *not* establish that per-sample gating works when the join is correct. Our companion study [11] examines that question directly and answers it in the negative for legal feedback delays; the relevant point here is that the two explanations for a null gate—no learnable signal, or a permuted label column—are not distinguishable by anything reported in a conventional table, and are distinguishable in one line by AOT.

7.6. The repair is not free

The obvious fix is to set the evaluation loader’s shuffle flag to **False**. It is the right fix, and it is not a silent one. A shuffling loader draws from the global pseudo-random stream once per epoch; removing those draws shifts every subsequent consumer of the stream—parameter initialisation of later components, dropout masks, and the training loader’s own shuffling—so the repaired run follows a different trajectory. Across 96 matched pairs that differ in exactly this flag, the reported test error changes by a median of

Table 13

Cost of the one-line repair: paired change in reported test MSE when the validation loader stops shuffling. Generated from `artifacts/rng_tax.csv`.

dataset	pairs	median $ \Delta $ (%)	max $ \Delta $ (%)	same best epoch (%)
ETTh1	24	0.013	0.497	79
ETTh2	24	0.019	2.749	88
ETTm1	24	0.012	0.493	46
ETTm2	24	0.078	4.316	46
all	96	0.019	4.316	65

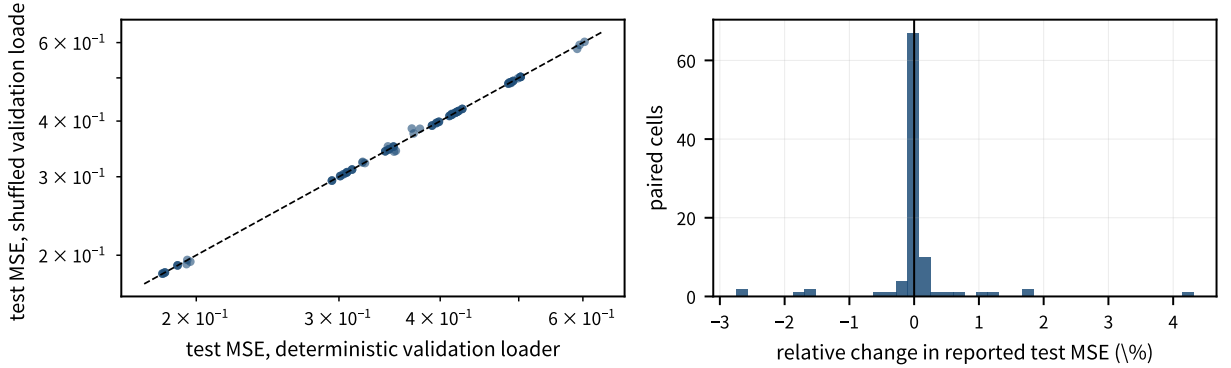


Figure 5: The repair changes the run, not just the artefact. Left: paired test MSE under the two loader settings. Right: distribution of the relative change. The change has no systematic direction; it is trajectory noise of the size that tooling-level nondeterminism is known to produce [30, 31].

0.019%, a ninetieth percentile of 0.837% and a maximum of 4.316%; 28.1% of pairs move by more than a tenth of a percent, only 0% are numerically identical, the selected epoch agrees in 64.6% of pairs, and a Wilcoxon signed-rank test finds no systematic direction ($p = 0.39$). The mechanism is visible in the split by early stopping: the 34 pairs in which the selected epoch differs move by a median of 0.1335%, an order of magnitude more than the 0.0114% of the pairs that stop at the same epoch, and the largest movement in the study belongs to the former group. The repair does not perturb the reported number directly; it perturbs the trajectory, and early stopping converts that into a visible change. Figure 5 and Table 13 show the distribution.

The direction of the change is uninteresting and the size is not: it is within the same order of magnitude as the margins that separate competing methods on these benchmarks, so it is large enough to move a comparison. The maintenance consequence is concrete. A team that discovers this fault in a published pipeline cannot patch the flag and keep the old table—the old table belongs to a different experiment. They must either re-run everything under the repaired configuration, or leave the flag alone and adopt the recorder of Section 5, which fixes the artefact without touching the random stream. We recommend the second for anything already published, and we note that this is a good reason for the contract to be a *recording* discipline rather than a change of loader behaviour.

8. Guidance

The findings compress into a short protocol. We state it as three roles because the checks that are cheap differ by role.

For authors (1) Never write a per-sample artefact without binding it to the identity of the samples it describes; the sidecar of Section 5—ordered sample-identifier digest, value digest, sample-set identity—is the minimum, and an order *label* alone is not sufficient. (2) Load per-sample arrays through a reader that

validates that record and fails closed, so a positional join cannot be written by accident; better still, join by sample identifier. (3) If a per-sample artefact must come from a shuffling loader, record the sample index alongside the value instead of disabling the shuffle—disabling it changes the experiment (Section 7.6). (4) Where the preconditions hold, run AOT on stored per-sample vectors as part of the test suite, at a conservative level, and read a pass as evidence against near-uniform scrambling only. Run CAT only if you can establish that the arms’ orders were produced independently; if the visiting permutations are not recorded, that cannot be established after the fact (Section 7.3). (5) When reporting a per-sample *null*, state explicitly which order check was performed; a shuffled-feature control is not one (Corollary 6), and if the analysis selects among several arms the control is optimistic rather than conservative (Proposition 7). (6) If the per-sample units overlap—sliding windows, nested segments, repeated measures—do not report significance from an independent-pairs formula; the error runs in the anti-conservative direction and can be large (Section 7.5).

For reviewers When a paper reports a per-sample analysis—difficulty prediction, per-instance routing, sample-level ablation—the question “how do you know the two tables are aligned?” has, at present, no standard answer, and the answers that sound convincing (a shape check, a shuffled-feature control, a sensible aggregate) are all provably uninformative here. Asking for the order check is cheap for an honest author and diagnostic otherwise. A reported null about per-sample structure deserves this question more than a reported effect does, because the fault only produces nulls. The useful follow-up is not “did you run a test” but “what does your artefact record about which sample each number belongs to”, because a passing diagnostic covers a narrower class of faults than its name suggests.

For artefact evaluators Badging schemes check that artefacts run and reproduce the reported numbers [32, 33, 34]. Both checks pass under this fault, by Proposition 2. A per-sample artefact can be checked further without the original authors and without a GPU: AOT reads the published vector, takes $O(n)$ time, and has an empirically calibrated false-alarm rate at the lengths we studied. We suggest that artefacts containing per-sample vectors carry sample-identity metadata as a condition of the reusable badge, and note that the check is mechanisable. We suggest metadata rather than a statistical test as the badge condition deliberately: the metadata check is complete for the faults it covers, whereas the test is not (Table 5).

9. Threats to validity and limitations

Construct Our diagnostics certify the presence of the structure that order induces; they do not certify alignment. AOT is invariant to circular rotation and nearly invariant to reversal, and—as Section 7.2 measures rather than conjectures—it did not flag any of the block permutations or the 5% and 10% partial permutations we injected. CAT is invariant to a permutation shared by all arms, and is defeated by *partial* sharing as well: a single shared pair among four arms was sufficient in every group we tested. An off-by-one write, a reversed sort, a batch-level shuffle or a reused loader seed therefore pass. We consider this a limitation of post-hoc auditing in general, not of these statistics in particular: content-based inference of order can only see structure that the transformation destroys. But we no longer describe the pair as covering the fault class, and the honest summary of our detector evaluation is that the covered region is narrow and that the real pipeline we audited sat mostly outside CAT’s part of it. The write-side contract is not subject to this limit, which is why we present it as primary—subject in turn to the 5 faults of Table 2 that it does not cover, of which the important one is two producers sharing a single wrong order.

Internal The observed defect is measured on our own pipeline, in which we deliberately reproduced the upstream loader policy. The corrupted conditions in T1 and T2 use permutations drawn the way a defective pipeline draws them—uniformly, and independently per arm—which matches the observed permutations of our shuffled runs but is an idealisation of any specific real defect. Our T2 gate is one reasonable design; a different gate would give different gains under the correct join. That does not affect the comparison of interest, which is between conditions holding the gate fixed.

External The empirical study is entirely within sliding-window multivariate forecasting on seven public benchmarks, three backbones and four arms. The formal results (Propositions 2–7) require only a per-sample artefact and a positional join, so they transfer verbatim to classification, ranking, retrieval or any other per-sample analysis; the *power* of AOT does not, since it is a function of the serial dependence that overlapping windows create. Section 7.4 quantifies exactly how that power decays, and CAT is the recommended instrument outside the sliding-window setting. We have not audited third-party artefacts, and we make no claim about the prevalence of the fault in the literature; establishing prevalence would require running the battery over public per-sample artefacts, which is mechanical, and which we regard as the natural next step rather than as part of this contribution.

Statistical The equivalence tests are paired over runs, and runs within a dataset are not independent; we therefore report intervals alongside verdicts, and where our sample cannot certify equivalence at the stated margin we say so rather than reporting a non-significant difference as an equality. AOT’s p -values are exact in the first two moments and asymptotic in the tail; the calibration in Section 7.2 bounds the discrepancy against a Monte-Carlo null at 0.0262, and any application that needs exactness can pay for the resampling.

10. Related work

Silent faults in machine-learning software Empirical studies of defects in deep-learning programs and frameworks consistently find that the most damaging class is the one that produces no crash [15, 14], and Tambon et al. [13] study exactly this class under the name *silent bugs*, cataloguing framework-level faults that alter results without raising an error. Our fault is of that kind but at a different level: it lives in the composition of two individually correct components, and its observable effect is confined to an artefact’s index. To our knowledge the per-sample order channel has not been analysed, and the interaction with negative controls has not been noted.

Nondeterminism and variance in training That a training run’s outcome moves under changes to random state and tooling is well documented [30, 31], and our Section 7.6 is an instance: the flag that causes the fault also couples the evaluation loader to the training trajectory. The contribution here is not the observation of variance but its consequence for repair—removing the shuffle is not a behaviour-preserving refactoring, so an honest fix is a re-run or a recording discipline, not a patch.

Leakage, reproducibility and evaluation pitfalls Kapoor and Narayanan [16] taxonomise leakage as the dominant reproducibility failure in machine-learning-based science, and reproducibility programmes and badging schemes formalise what an artefact must contain [33, 34, 32]. Leakage inflates results and is caught by out-of-sample discipline; the fault we study deflates them and is caught by none of the standard checks, which is why we treat it as a distinct failure mode rather than a variant. In time series specifically, evaluation methodology has been scrutinised for split construction and dependence between folds [18, 1, 35]; that literature concerns *which* samples are used, whereas our concern is the identity of a sample once used.

Provenance and workflow systems Recording how a derived artefact came about is the subject of a mature literature [22, 23], is operationalised by workflow engines [24, 25] and experiment trackers [26], and is codified for scientific data by the FAIR principles [36]. Our contract is a deliberately minimal specialisation: for the single property that positional joins depend on, a digest over the ordered sample identifiers, a closed vocabulary of order semantics, and a reader that fails closed. We see it as an instance of the general prescription that the ML tooling stack currently omits, not as a substitute for a workflow engine.

Testing without an oracle Metamorphic testing [37, 38] verifies programs by checking relations between outputs of related inputs when no oracle is available. AOT and CAT are statistical relatives: neither knows the correct order, and each checks a property that the correct order implies—serial dependence for AOT, cross-arm agreement for CAT. The connection is practically useful, since it places both tests naturally in a test suite rather than in an analysis notebook.

Negative controls Negative controls are a standard device for detecting bias in observational studies [17], and their machine-learning analogues—shuffled labels, shuffled features, random baselines—are used to validate per-sample claims. Proposition 5 identifies a case in which a negative control is provably uninformative, because the fault it is meant to detect and the control itself induce the same distribution. We are not aware of a prior statement of this degeneracy, and we suspect it is not unique to order corruption: any fault that acts by destroying a pairing will collide with the control that destroys the same pairing. That suggests a general design rule, which we state as a question for future work: a negative control is only informative about faults that act differently from the control.

Time series forecasting and plug-ins The empirical carrier of this study is the long-horizon forecasting benchmark suite [6, 7, 8, 9, 10] and the model-agnostic modules attached to it [27, 29, 28], implemented in a common reference library [4, 5]. We use them only as a source of realistic per-sample artefacts, and we take no position here on the arms’ relative merits; the companion study [11] addresses the selection question.

11. Conclusion

Per-sample analysis rests on an assumption that is almost never written down: that two artefacts produced by different parts of a pipeline are in the same order. We have shown that when the assumption fails, the failure survives the guards a pipeline conventionally applies—the sample-weighted aggregate metric is invariant, the length check is blind by construction, and, most importantly, the negative control that is supposed to protect the analysis is *equal in distribution* to the fault it is meant to detect. A null result about per-sample structure, presented with a well-behaved shuffled-feature control, is therefore exactly as consistent with a broken join as with a true null, and nothing in a conventional table can separate the two. The one class of analyses where the control does behave differently—those whose label is derived from several independently written vectors—is the class where it is most misleading, because there it reports a milder failure than the fault produces.

The situation is repairable, but not by the route we first proposed. Write-side is where the leverage is: bind every per-sample artefact to the ordered identity of the samples it describes—not merely to a label naming its order—and refuse to load one for a positional join without verifying that binding. On 25 injected identity faults the contract we specify stops 20, including the rotations, permutations, stale sidecars and equal-length substitutions that no content-based test can see, and we enumerate the 5 it does not stop.

Read-side, for artefacts that already exist, the two diagnostics are worth running but cover less than their names suggest, and reporting that boundary is the more useful of our two empirical contributions. On 356 runs AOT certifies 100% of intact sequences and 6.18% of uniformly permuted ones at $\alpha = 0.05$ and separates the two populations perfectly at $\alpha = 10^{-6}$; it flags 90.63% of arms on the real defect. But it flagged 0% of the injected block permutations and 0% of the mild partial permutations, so it is a diagnostic for near-random scrambling and not an integrity test. CAT is sharper still in its dependence on an assumption we had not checked: it requires each arm’s order to have been drawn independently, and in our own archive 20 of 24 defective groups share an order between arms, whereupon CAT certifies 91.67% of genuinely corrupted groups as clean. A single shared pair among four arms defeats it. We also found that the obvious one-line repair silently changes reported numbers by up to 4.316%, so a maintainer must re-run rather than patch—or adopt the recorder, which fixes the artefact and leaves the run alone.

Three directions follow. The first is mechanical and, we think, worth doing: run the diagnostics over the per-sample artefacts that accompany published papers and measure how often the fault is present in the wild, remembering that a non-certification is evidence of near-random scrambling rather than of order integrity in general. The second is to close the diagnostic gap we have documented: AOT’s insensitivity to block and mild partial permutations, and CAT’s dependence on independent arm orders, are both invitations to design statistics that target locality-preserving corruption and correlated loader state. The third is conceptual. Our degeneracy result is an instance of a pattern—a control that destroys the same structure as the fault cannot see the fault—and we expect other instances to exist wherever a validation device and a failure mode act through the same mechanism. Cataloguing them would tell us which of the checks we currently trust are load-bearing.

CRediT authorship contribution statement

Mianhan Liu: Conceptualization, Methodology, Software, Formal analysis, Investigation, Data curation, Visualization, Writing – original draft, Writing – review & editing. **Chen Chen:** Validation, Investigation, Software, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used a large language model to assist with copy-editing, LaTeX formatting and the drafting of analysis and plotting scripts. All experimental design, statistical decisions, numerical results and scientific conclusions were produced and verified by the authors, who take full responsibility for the content of the publication.

Relation to other work by the authors

This study and the companion manuscript [11] share an experimental infrastructure and a research programme but address disjoint questions and report disjoint results: the companion asks whether per-window plug-in selection is learnable under legal feedback delay, and this paper asks whether a per-sample artefact is in the order its consumer assumes. To let editors and reviewers assess the overlap directly we state it explicitly. *Shared:* the forecasting grid and the trained checkpoints that produced the per-window error vectors, the dataset preparation code, and the two bug fixes reported in Section 7.3 (the loader-order defect and the corrected cross-arm variance term), which are reported identically in both manuscripts. *Exclusive to this paper:* the degeneracy proposition and its controls, the AOT and CAT diagnostics with their calibration, sensitivity and blind-spot analyses, the observed-permutation audit of the archive, the provenance contract and its fault-injection coverage, the dependence-aware re-analysis of per-window difficulty correlations, and the repair-cost measurement. *Exclusive to the companion:* everything concerning selector training, feedback delay, oracle headroom and gating policy. No contribution is claimed in both manuscripts, and neither manuscript’s conclusions depend on the other being accepted. The two releases are versioned and archived independently, and the numbers quoted across them are produced by the same verification scripts.

Vitae

Mianhan Liu works on empirical evaluation methodology for machine-learning systems, with an emphasis on reproducibility, silent data-handling faults and the statistical validity of per-sample analyses. He designed the study, implemented the audit tools and the experimental grid, and wrote the manuscript.

Chen Chen works on applied machine learning and experiment tooling. He independently replicated the released artefacts and the verification pipeline, reviewed the statistical procedures and contributed to the manuscript.

Data availability

All datasets used in this study are public standard benchmarks (ETT, Weather, Exchange-Rate, ILI) and are not redistributed here; the release includes the download script and the checksums we verified. Everything required to recompute every number, table and figure in this manuscript is released at <https://github.com/mliu1994/CRediT>.

[//github.com/oscarliu2019/order-provenance-audit](https://github.com/oscarliu2019/order-provenance-audit): the experiment configuration; the per-run table for all 356 runs; the per-sample audit outputs (AOT sequence statistics, calibration, sensitivity, certification sweep, cross-arm agreement, the battery); the recovered permutations of the 24 defective groups and the controlled sharing conditions used to probe CAT's independence assumption; the per-sample feature tables; the T1 and T2 outputs, the dependence-aware and cluster-bootstrap re-analyses and the purge audit; the equivalence tests; the paired repair-cost table; the reference implementation of the provenance contract, its 25 fault-injection cases and both diagnostics; the test suite covering the propositions as executable statements; and the generators for `paper/numbers.tex`, all tables and all figures. Per-window error vectors are released as arrays together with their sidecars, so that every derived per-sample conclusion can be recomputed rather than merely inspected. A verification script, `tools/verify_paper_numbers.py`, re-derives every quantitative claim in this manuscript from those artefacts and fails on any mismatch, printing the claim identifier, the printed value and the recomputed value.

The repository is also archived permanently in Software Heritage: the revision from which this manuscript was produced is `swh:1:rev:1094bb72d608638a694b763c36a8f7ee875a8111` (origin <https://github.com/oscarliu2019/order-provenance-audit>, visit `swh:1:snp:ad97da379ee5cbf09a196bad6883fed9926cbaac`).

The exact release from which every number in this manuscript was computed is archived, with all artefacts, at <https://doi.org/10.5281/zenodo.22809148> (DOI: 10.5281/zenodo.22809148); the repository above is its development mirror.

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